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Explainable AI machine learning framework for chronic kidney disease prediction utilizing electronic health records

Muhammad Rizwan[1†], Rashid Naseem[1†], Muhammad Ahmad Khan[1,2], Arshad Ahmad[3], Muhammad Fozan[4] and Nazeer Muhammad[5*]

Abstract

Background Chronic Kidney Disease (CKD) is one of the significant health issues in the world which is usually unnoticeable and afterwards causes greater morbidity and mortality. Timely intervention is important in early diagnosis that will enhance patient outcomes.

Methods This study proposes an explainable artificial intelligence driven machine learning (ML) framework for CKD prediction using electronic health records. A dataset of 398 patients (241 CKD and 157 non-CKD) from Pakistan Kidney Center, Abbottabad, was analyzed using multiple models, including logistic regression, decision tree, k-nearest neighbors, gradient boosting, CatBoost, AdaBoost, XGBoost, and random forest (RF). Feature selection methods mutual information, DT, and RF identified the most informative clinical variables, and experiments were conducted using both full and top 10 feature subsets. Model performance was evaluated using stratified 5-fold and 10-fold cross-validation with metrics such as accuracy, precision, recall, F1-score, balanced accuracy, MCC, and AUC. Statistical validation was performed using confidence intervals, McNemar's test, ANOVA, and Tukey HSD. Model interpretability was achieved using Hapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME).

Results The results demonstrated consistently high performance across all models, with the RF model achieving an accuracy of 98%. Statistical analysis revealed no significant differences among top-performing models, indicating comparable predictive capabilities. Feature-level analysis identified key biomarkers—estimated glomerular filtration rate, creatinine, and urea—as having large effect sizes and strong discriminative power. The reduced feature set maintained competitive performance, highlighting the efficiency of selected variables. Model interpretability using SHAP and LIME provided transparent and clinically meaningful insights, confirming the importance of these features in CKD prediction.

Conclusion The proposed framework demonstrates robust and reliable performance for CKD prediction, supported by comprehensive statistical validation and explainability analysis. The findings indicate that predictive performance

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is primarily driven by the strength of selected clinical features, enabling multiple ML models to achieve comparable results. This approach offers a practical, interpretable, and generalizable solution for early CKD detection in clinical settings.

Keywords Chronic kidney disease, Machine learning, EXplainable artificial intelligence, Electronic health records, Random forest

Introduction

Chronic kidney disease (CKD) is a significant epidemic affecting the global community. Although its effects are becoming increasingly well-known, the public awareness of CKD remains low [1]. Approximately 674 million individuals (approximately 8.5% of the global population) are estimated to have the CKD. It is the most rapidly increasing causes of premature mortality and is expected to be the fifth leading cause of years of lost life by 2040 [2]. It is further observed that kidney dysfunction increases the risks of cardiovascular diseases while complicating the management of diabetes and hypertension. Furthermore, it is also associated with cancer and dementia [3].

Rates of mortality before 65 years of age are of special concern because a considerable proportion of individuals with CKD develop end-stage renal failure (ESRF). This progression severely reduces the quality of life and leaves a significant economic burden on society and the economy [4, 5]. Still, 10% who require dialysis or transplantation do so, at a cost that occupies up to 6.7% of medicare in the U.S., as much as 26% of healthcare expenditure in other countries. These statistics highlight the high priority of repositioning CKD as a health concern worldwide as an urgent need to address it is essential [3].

Although there is a considerable disparity in the incidence of renal replacement therapy (RRT) in different countries, the lowest-income countries having fewer than 50 cases per million people. While the highest-income countries having 150–400 cases per million people but the healthcare systems do not meet the increases in demand [6, 7]. Individuals over 60, smokers and drug users are more at risk due to damage to the kidney tissues and blood supply as depicted in Fig. 1. Other conditions include glomerulonephritis, nephrolithiasis, blood clots, as well as some rare diseases that can lead to death [8, 9]. CKD can also be silent at early stages, which makes it difficult to diagnose it in an early phase of the illness as well as complicates detecting it during its acute phases [10–12]. In Pakistan, CKD is facilitated by unhealthy diet, self-medication, contaminated water, hypertension, diabetes and lack of awareness is highly prevalent [13–15]. Factors highlight the necessity of early detection and lifestyle intervention [16, 17].

Screening of patient at the early stages of CKD can substantially help to avoid the development of end-stage kidney disease (ESKD) [18, 19]. Early diagnosis helps in timely intervention so that the nature of the disease can be changed and reducing the risk of CKD complications. Aspiring diagnostic tools of kidney biopsies are useful for evaluating both pathological and functional impairment of affected persons [20]. In clinical practice, a reduced estimated glomerular filtration rate (eGFR) below 60 mL/min/1.73 m^2, along with an elevated protein-to-urine-volume ratio, indicates impaired kidney function. Moreover, individuals with CKD face a heightened risk of death from cardiovascular conditions, particularly heart attacks [21].

Medical professionals are able to halt the progression of renal function or failure with the use of early stage prediction and diagnosis [22]. Creatinine level in human blood and urine is used to evaluate eGFR value, which is an important non-invasive event in renal function and an evolutionary risk factor. When it comes to challenging aspects of the kidney's function, eGFR fails to produce good results and has a number of limitations, such as creatinine and non-imposing delegates [23]. One of the most important uses of the intelligent structure is in the medical industry. Data about the patient's symptoms and drugs is frequently used by medical professionals to make better treatment decisions [24–27].

Use of artificial intelligence (AI) improves productivity and shortens the time to predict the CKD [28–30]. In this instance, a nephrologist can analyze the test reports generated by the AI processing, as seen in Fig. 2. An accurate diagnosis is provided by the nephrologist after reviewing the results given by the AI. Patients benefit from a speedy diagnosis since it allows for the determination and implementation of treatment plans more rapidly [31, 32]. Finally, because decisions are made more quickly and with fewer mistakes, it leads to greater patient health. Researchers and practitioners are using machine learning (ML) (an AI subfield) that has started to make waves in the healthcare sector due to the types of data it can process and the results it produces [33]. Researchers developed deferent ML models for the prediction of CKD. These models use information obtained from the test results (e.g., creatinine, eGFR, Urea, and Hemoglobin etc.) of the patients [34, 35].

Here is how the rest of this paper is structured: In the section "Literature review" literature review, we presented the summaries of previous studies that have used ML and XAI to predict CKD. Section "Contribution" highlights the contributions of this research. Section "Proposed framework" presents the methodology of the

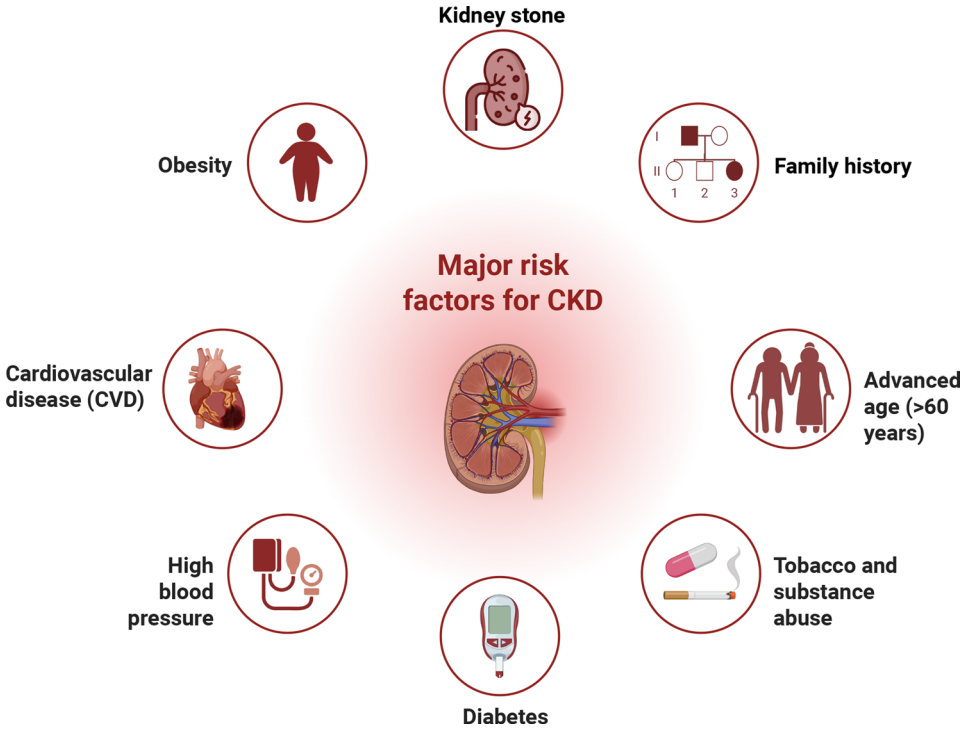


Fig. 1 Major clinical and lifestyle risk factors for CKD, including diabetes, hypertension, cardiovascular disease, obesity, advanced age, family history, kidney stones, and tobacco/substance use

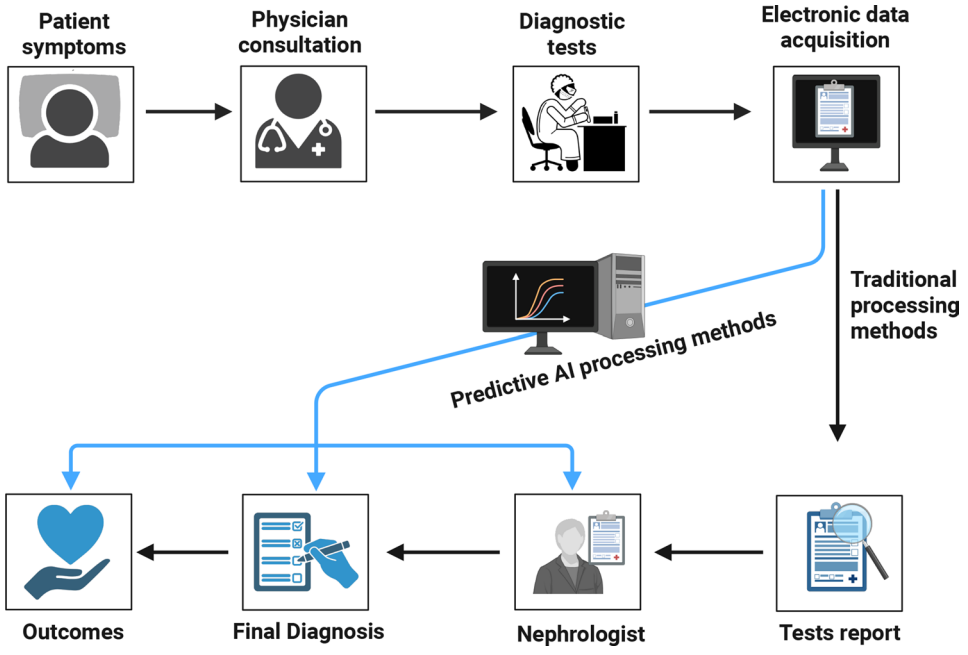


Fig. 2 Diagnostic workflow: patient presentation prompts physician consultation and medical tests, leading to electronic data acquisition. Data can be analyzed using traditional statistical approaches or predictive AI-based models. AI enables rapid, automated CKD risk prediction, generating diagnostic reports that assist nephrologists in clinical decision-making and improving patient outcomes

proposed approach. Section "Results and discussion" provides an analysis of the experimental findings, and finally, section "Conclusion" summarizes the paper.

Literature review

Researchers, Çelik et al. [36] have utilized diverse supervised ML classifiers to predict CKD, frequently leveraging the UCI-ML dataset [37] sourced from Apollo Hospitals. Studies comparing traditional algorithms found that Decision Trees (DT) often outperformed Support Vector Machines (SVM), while Random Forest (RF) and Extra Tree (ET) classifiers also yielded superior results in multiple experimental evaluations [36, 38, 39]. Furthermore, Devika et al. [40] employed several classification algorithms, including RF, naïve bayes (NB), and k-nearest neighbors (KNN), in their research. Concurrently, Rady and Anwar in 2019 [7] used SVM, probabilistic neural networks (PNN), multilayer perceptrons (MLP), and radial basis function (RBF) networks to classify CKD.

Khan et al. [41, 42] conducted experiments on UCI-ML [37] using seven ML algorithms such as NB tree (NBT), logistic regression (LR), SVM, and MLP, NB, and clustered hierarchical iterative reparameterization and projection (CHIRP). CHIRP, reached a great extent of accuracy. Parallel to these findings, Ghosh et al. [43] used four viable algorithms. SVM, adaptive boosting (AdaBoost), Linear Discriminant Analysis (LDA), and gradient boosting. Gradient Boosting imposed the best prediction accuracy.

Beyond standard supervised learning, in 2019, Elhoseny et al. [44] applied ant colony optimization (ACO) with density-based features (DFS) for CKD classification due to their reduced features and improved prediction rates. In 2021, investigation into unsupervised models revealed that k-means clustering provides more accurate preliminary predictions compared to methods like autoencoder, clustering, DB-Scan, and isolation forest (I-Forest) [45].

Article by Krishnamurthy et al. [46] used data collected from the Taiwan National Health Insurance Research Database (NHIRD) [47] to classify the CKD using ML techniques i.e., DT, RF, LR, LightGBM, CNN, and BDLSTM. Authors reported, the better experimental results for CNN.

Recent literature also emphasizes the integration of XAI tools, such as LIME and SHAP, to enhance clinical decision-making and ensure healthcare compliance by making AI outputs understandable to medical experts [48]. Subsequently, researchers developed an XAI model for early CKD diagnosis using the SCI-XAI framework to balance high accuracy with clinical transparency [49]. Model uses extreme gradient boosting (XGBoost) classifier and produced better results with reduced number of features. To make these models practical for healthcare, XAI strategies such as LIME and SHAP were implemented to make AI decisions transparent for medical experts [50, 51].

Comparative analysis

According to the discussion in Section 2, ML strategies have successfully demonstrated remarkable outcomes. However, despite of numerous ML approaches published in the literature, the precision and efficacy of ML models for CKD are significantly impacted by variables such as the training dataset, data preparation, and input data. Discussion in this section has also led to the conclusion that the UCI-ML repository dataset is the primary focus of numerous research projects, despite the fact that other datasets have also been used for CKD prediction. In addition, CatBoost, AdaBoost, XGBoost, RF, LR, DT, KNN, Gradient Boosting, and SVM are among the ML algorithms that were employed to predict CKD.

Additionally, the accuracy, precision, recall, F1-score, and the area under the ROC curve (AUC-ROC) are important evaluation metrics for evaluating the performance of ML algorithms. While, XAI techniques such as SHAP and LIME have been introduced as a result of recent improvements in model interpretability.

Literature that has been discussed (see Section "2") is summarized and presented in the comparative Table 1. In this table, the second column contains the authors and year, while the third column displays the data sets that were used in the experiments. Fourth column lists the ML techniques used. Evaluation parameters are mentioned in the fifth column. Similarly, columns six and seven display the XAI methods of the investigation, which include SHAP and LIME.

Contribution

ML has shown better accuracy in the classification of CKD, utilizing the data from the sources such as UCI ML-dataset repository, as well as CKD-specific cohorts [54, 55]. Nevertheless, according to previous studies, it is necessary to consider more detailed healthcare data when exploring and creating ML models [52, 56]. Unlike existing CKD prediction studies, to the best of our knowledge, that rely on single-model evaluation, this work proposes an XAI based ML and statistically validated framework combining multi-criteria decision analysis in newly collected dataset from Pakistan Kidney Center (PKC), Abbottabad, KPK, Pakistan.

Key contributions of this work are as follows:

- Development and validation of an XAI based ML framework for CKD prediction using clinical data;
- Integration of advanced statistical validation techniques, including confidence intervals,

Table 1 Comparative analysis of CKD-related literature

Sr. No	Authors & Year	Dataset Used	ML Algorithms	Performance Metrics	XAI (SHAP)	XAI (LIME)	TOPSIS
1	Celik et al., 2016 [36]	UCI-ML [37]	SVM, *DT*	PCC, Kappa, MAE, MSE, RAE, RSE, ARRSE	×	×	×
2	Devika et al., 2019 [40]	UCI-ML	KNN, NB, *RF*	ACC, PRE, FME, REC, CM	×	×	×
3	Rady & Anwar, 2019 [7]	UCI-ML	*PNN*, MLP, SVM, RBF	Sensitivity, Specificity, ACC	×	×	×
4	Elhoseny et al., 2019 [44]	UCI-ML	*D-ACO*	ACC, FME, Kappa, Sensitivity, Specificity, FPR, FNR	×	×	×
5	Pasadana et al., 2019 [39]	UCI-ML	DS, HT, J48, CTC, J48graft, LMT, NBT, *RF*, RT, REPT, SC	ACC, PRE, REC, MAE, FME, Kappa, Runtime	×	×	×
6	Ekanayake & Herath, 2020 [38]	UCI-ML	DT, *RF*, XGB, *ET*, AdaBoost, KNN, CNN, SVC Linear, LR, SCV, RBF, GNB	PRE, REC, FME	×	×	×
7	Khan et al., 2020 [41]	UCI-ML	NB, LR, MLP, J48, SVM, NBTree, *CHRIP*	ACC, PRE, FME, REC, MAE, RAE, RMSE, RRSE, ROC, AUC	×	×	×
8	P. Ghosh et al., 2020 [43]	UCI-ML	SVM, AdaBoost, LDA, *Gradient Boosting*	Sensitivity, Specificity, ACC, PRE, FME, MAE, MSE, RMSE, AUROC	×	×	×
9	Krishnamurthy et al., 2021 [46]	Taiwan's NHIRD [47]	*CNN*, BLSTM, LightGBM, LR, RF, DT	Specificity, ACC, PRE, REC, FME, AUROC	×	×	×
10	Antony et al., 2021 [45]	UCI-ML	*K-Means*, DB Scan, Autoencoder, Iforest	ACC, PRE, REC, FME, RI, MI, V-measure	✓	×	×
11	Ebiaredoh-Mienye et al., 2022 [48]	UCI-ML	LR, DT, RF, SVM, *XGBoost*, AdaBoost, CS AdaBoost	ACC, AUROC, Sensitivity, Specificity	×	×	×
12	Moreno-Sanchez, 2023 [49]	UCI-ML	*XGBoost*, RF, RT, AdaBoost	ACC, PRE, REC, FME, Specificity, Interpretability, Fidelity, FII, FAI	✓	×	×
13	Arif et al., 2024 [52]	UCI-ML	Ridge, SGD, Bernoulli NB, LR, RF, DT, Gaussian NB, *MLP*	ACC, PRE, REC, FME, ROC, AUC	×	✓	×
14	**This Study**	PKC dataset [53]	LR, DT, KNN, Gradient Boosting, XGB, SVM, CatBoost, AdaBoost, *RF*	ACC, PRE, REC, FME, Balanced Accuracy, MCC, Specificity, AUROC	✓	✓	✓

*In the fourth column, the underlined and *italic* ML algorithms achieved the highest accuracy within its respective study;*Abbreviations used in the table are defined in the supplementary document S1

McNemar's test, ANOVA and Tukey HSD Post-hoc Test to provide rigorous and reliable comparison of model performance;

- Application of multiple feature selection methods (Mutual Information, DT, and RF importance) to identify clinically relevant features and evaluate model performance on both full and reduced feature sets;
- Introduction of a multi-criteria decision-making approach (TOPSIS) to rank ML models based on multiple performance metrics rather than relying on a single evaluation criterion;
- Incorporation of XAI techniques (SHAP and LIME) to enhance interpretability, providing both global and local explanations aligned with clinical knowledge of CKD indicators;
- Extensive performance evaluation using a wide range of metrics, including Accuracy, Precision, Recall, F1-score, Balanced Accuracy, Specificity, MCC, ROC-AUC, PR-AUC, and calibration analysis, ensuring clinical reliability of predictions.

Proposed framework

This study proposes an XAI enabled ML framework for CKD prediction. Overall workflow is illustrated in Fig. 3, which includes data preprocessing, feature selection, model development, evaluation, and interpretability. Multiple ML models are trained and evaluated using both train–test split and cross-validation strategies. Model performance is assessed using various evaluation metrics and statistical tests, while SHAP and LIME are applied to provide interpretability. The complete procedure is summarized in Algorithm 1, and detailed steps are described in the following subsections.

Data acquisition and description

Dataset used in this research, was collected from PKC during August and October 2023. Comprising records from both healthy individuals and patients diagnosed with CKD [53]. Clinical and laboratory parameters were recorded during patients visits to PKC. The dataset represents a curated clinical dataset derived from real-world

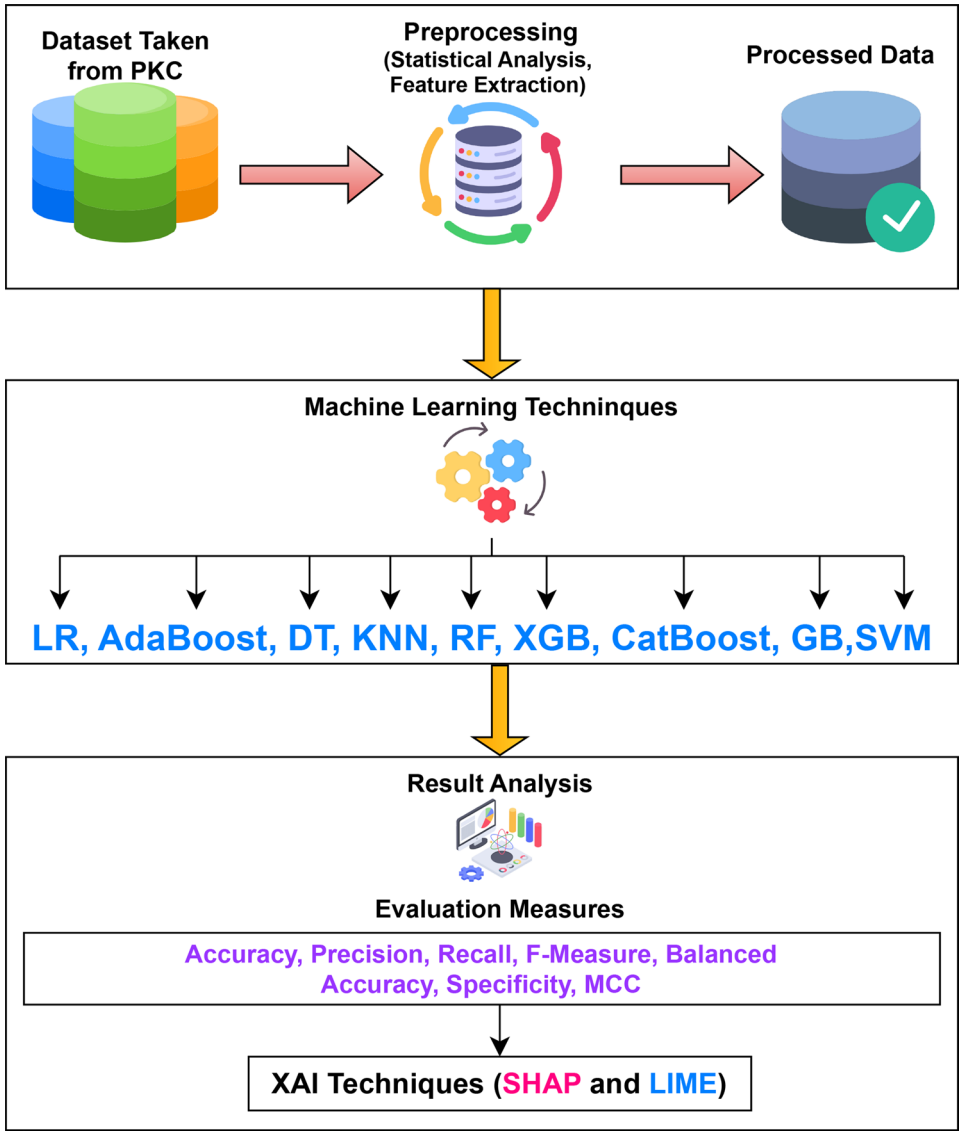


Fig. 3 Workflow for the proposed framework

patient records, ensuring its relevance for practical healthcare applications.

No informed permission was necessary because all patient data was anonymised and de-identified before analysis. All experiments were conducted in accordance with approved protocols and in alignment with the ethical principles of the Declaration of Helsinki. Exclusion and inclusion criteria, as well as the dataset collection flow design, are displayed in Fig. 4.

Dataset includes 22 variables, encompassing demographic data (e.g., age, sex), laboratory test results (e.g., creatinine, urea, haemoglobin, eGFR), and clinical observations. Laboratory values were measured using standardized diagnostic equipment under the supervision of hospital laboratory technicians, following PKC's quality assurance protocols. The target label was assigned based on a nephrologist-confirmed diagnosis of CKD. All data were digitized using a secure hospital management system. Identifying information was removed to ensure full anonymization before data sharing. Out of a total of 398 entries, 241 were identified as CKD cases, while the remaining 157 were non-CKD. 60.55% are CKD instances, whereas 39.45% are non-CKD cases. There are people in the dataset whose ages vary from six to

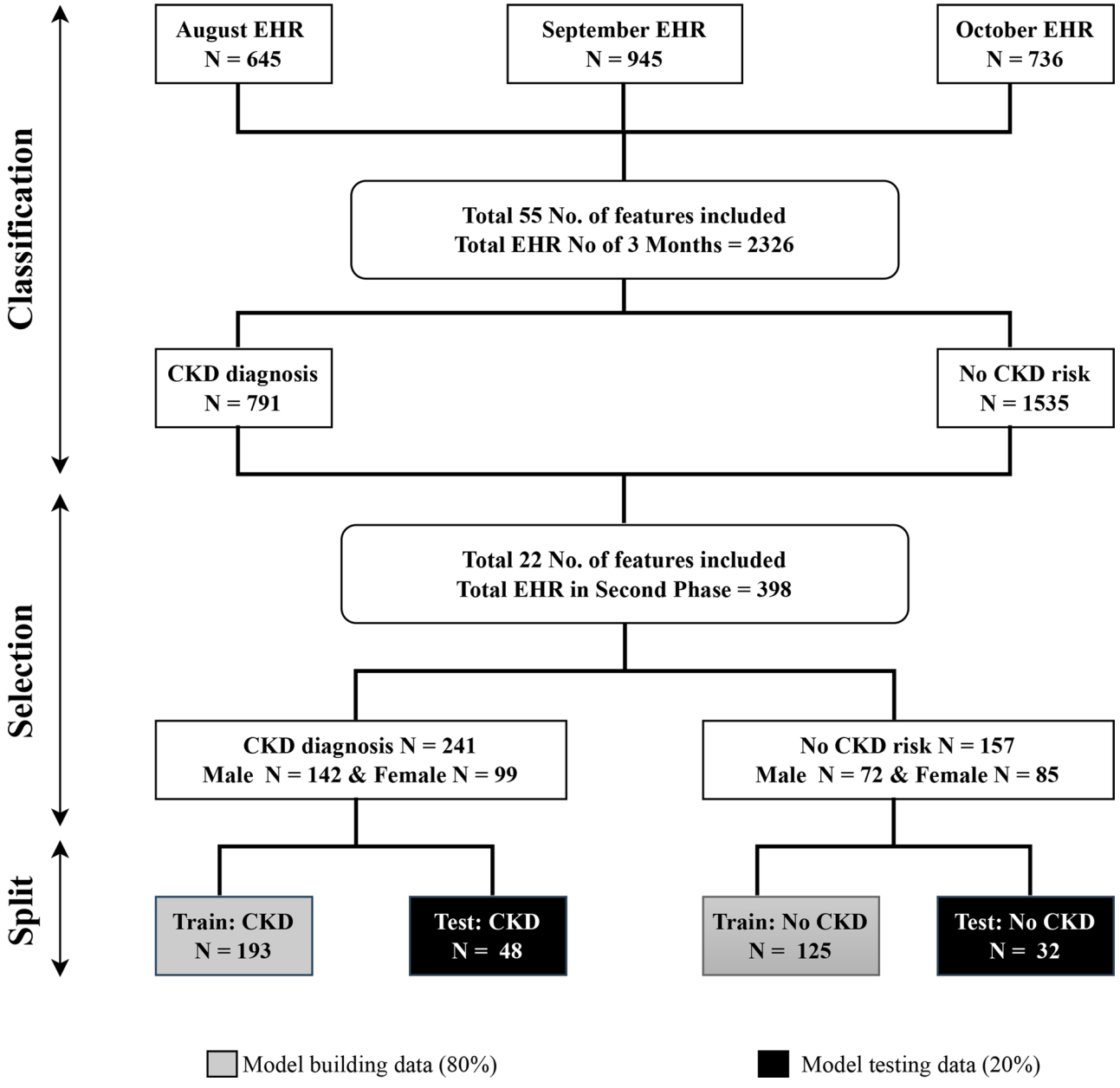


Fig. 4 Summary of patient data inclusion and exclusion applied in the dataset

ninety-eight. With 20 numerical features and 2 nominal characteristics, the CKD dataset has a total of 22 features. Table 2 further shows the details of dataset which used to classify the CKD.

Data preprocessing

Data preprocessing was performed to ensure data quality and improve model performance. Initially, the dataset was examined for inconsistencies and prepared for analysis. Categorical variables, such as gender, were encoded into numerical form to make them suitable for ML models. Since the dataset consists of numerical clinical features, normalization was applied where necessary to ensure consistent feature scaling, particularly for distance-based models. The dataset was then divided into training (80%) and testing (20%) sets using stratified sampling to preserve the original class distribution. This resulted in 193 CKD and 125 non-CKD samples in the training set, and 48 CKD and 32 non-CKD samples in the testing set. This preprocessing step ensures that the data is appropriately structured and balanced for subsequent modeling and evaluation.

Table 2 Feature-wise descriptive statistical analysis of the CKD dataset, including variable definitions, measurement units, unique values, and data distribution

Sr. No	Features	Abbrev.	Domain	Data Type	Units	Unique	Description
1	Sex (male (1): female (0))	Sex	Demographic	Nominal	Male/Female	2	Gender of the patients
2	Age	Age	Demographic	Numerical	Years	71	Age of the patients in years
3	Hemoglobin	Hb	Clinical Data	Numerical	g/dL	83	Hemoglobin levels in the blood
4	Red Blood Cell Count	RBC	Clinical Data	Numerical	$\times 10^6/\mu L$	44	Number of RBC per microliter
5	White Blood Cell Count	WBC	Clinical Data	Numerical	$\times 10^3/\mu L$	110	Number of white blood cells per microliter
6	Hematocrit	Hct	Clinical Data	Numerical	%	211	Percentage of red blood cells in blood
7	Lymphocytes	Lymphs	Clinical Data	Numerical	%	47	Percentage of lymphocytes
8	Monocytes	Monos	Clinical Data	Numerical	%	11	Percentage of monocytes
9	Eosinophils	Eos	Clinical Data	Numerical	%	10	Percentage of eosinophils
10	Platelet Count	PLT	Clinical Data	Numerical	$\times 10^9/L$	205	Platelet count per liter
11	Mean Corpuscular Hemoglobin	MCH	Prognosis & Complications	Numerical	pg	81	Hemoglobin concentration in grams per deciliter
12	Mean Corpuscular Hemoglobin Concentration	MCHC	Prognosis & Complications	Numerical	g/dL	70	Hemoglobin concentration
13	Mean Corpuscular Volume	MCV	Prognosis & Complications	Numerical	fL	84	Average RBC volume
14	Mean Platelet Volume	MPV	Prognosis & Complications	Numerical	fL	72	Average platelet size
15	Plateletcrit	PCT	Prognosis & Complications	Numerical	%	44	Platelet volume percentage
16	Platelet Distribution Width	PDW	Prognosis & Complications	Numerical	%	104	Platelet size variability
17	Polymorphs	PMNs	Prognosis & Complications	Numerical	%	52	Neutrophil percentage
18	Red Cell Distribution Width SD	RDW-SD	Prognosis & Complications	Numerical	%	136	RBC size variation
19	Creatinine	Cr	Biochemical Variables	Numerical	mg/dL	128	Creatinine level in blood
20	Urea	Urea	Biochemical Variables	Numerical	mg/dL	172	Urea level in blood
21	Glomerular Filtration Rate	eGFR	Biochemical Variables	Numerical	mL/min	346	Kidney filtration rate
22	Class (CKD (1): Non-CKD (0))	Class	Associated Disease	Nominal	CKD/Non-CKD	2	CKD diagnosis indicator

Algorithm 1 Proposed Framework for CKD Prediction Using ML, XAI, and TOPSIS

Let $S = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$ be the clinical dataset.
Perform data preprocessing (cleaning, encoding, normalization).
Apply stratified split: $S \rightarrow S_{\text{train}}$ (80%), S_{test} (20%).
Perform exploratory and statistical analysis on S.
Apply feature selection methods:
  Compute Mutual Information, Decision Tree, and Random Forest importance.
  Select top-k features form reduced dataset S^*.
Define machine learning models:
  ML_models = {LR, DT, KNN, Gradient Boosting, CatBoost, AdaBoost, XGBoost, RF, SVM}.
for each model M in ML_models **do**
    Perform hyperparameter tuning using GridSearchCV.
    Train M on S_{train}.
    Evaluate M using:
      Accuracy, Precision, Recall, F1-score,
      Specificity, Balanced Accuracy, MCC, ROC-AUC, PR-AUC.
    Perform cross-validation (k = 5, 10).
end for
Perform statistical validation:
  Compute Confidence Intervals.
  Apply McNemar's test, ANOVA and Tukey HSD Post-hoc Test.
Rank models using TOPSIS based on multiple evaluation metrics.
Select best model M_{best}.
Apply eXplainable AI techniques:
  Compute SHAP values for global and local interpretation.
  Apply LIME for instance-level explanations.
Analyze feature contributions and validate clinical relevance.
return final model M_{best} with explanations and ranking.

Exploratory and statistical analysis

An exploratory and statistical analysis was conducted to understand the distribution and clinical relevance of the dataset features. Descriptive statistics, including mean, standard deviation, minimum, and maximum values, were computed for all numerical attributes to summarize their characteristics.

Analysis revealed that key clinical indicators such as creatinine, urea, and glomerular filtration rate (GFR) exhibit significant variation between CKD and non-CKD patients, highlighting their importance in disease prediction. A summary of the descriptive statistics is presented in Table 3, which provides an overview of the dataset features and their distributions. These insights guided subsequent feature selection and model development.

Feature selection

Feature selection was performed to identify the most relevant clinical attributes for CKD prediction and to reduce model complexity. Three techniques were employed: Mutual Information (MI), DT feature importance, and RF feature importance. MI was used to measure the dependency between each feature and the target variable, while tree-based methods (DT and RF) were utilized to evaluate feature importance based on their contribution to reducing impurity during model training. Based on the combined insights from these methods, the most informative features, including key clinical indicators such as creatinine, GFR, and urea, were selected for subsequent modeling

Machine learning models

Different ML algorithms, RF, LR, DT, KNN, Gradient Boosting, CatBoost, AdaBoost, SVM, and XGBoost, were implemented to predict CKD. To interpret the model's predictions, SHAP was applied, providing insights into feature contributions at both global and local levels. For comparative analysis of feature attributions, LIME was also used to generate localized interpretability. Structure of the proposed explainable ML framework is shown in detail in the algorithm.

LR was used as a baseline linear model, while DT provided a simple and interpretable tree-based approach. RF and Gradient Boosting methods were included as ensemble techniques to improve predictive performance and reduce overfitting. Support Vector Machine and KNN were applied as representative distance-based and kernel-based classifiers. Additionally, advanced boosting algorithms such as XGBoost and CatBoost were incorporated due to their effectiveness in handling structured tabular data.

These models were selected to provide a comprehensive comparison across different learning paradigms, including linear, tree-based, ensemble, and kernel-based methods. This diverse selection ensures a robust evaluation of model performance across different classification strategies.

Table 3 Descriptive statistics of CKD dataset features

Feature	Mean	Std Dev	Min	Max	25th %	50th %	75th %
Sex	0.54	0.50	0	1	0	1	1
Age	49.34	16.58	6	98	36	49	63
Haemoglobin	11.07	1.87	6.1	16.1	9.8	11	12.2
Hematocrit	32.15	6.01	18	53.2	28.1	31.59	35.81
Lymphocytes	22.64	9.46	2	71	16	22	28
MCH	29.28	3.09	19	39.2	28	29.2	31
MCHC	34.70	1.87	27	42	34	35	36
MCV	84.95	9.18	55	122	80	85	90.93
Monocytes	4.61	2.03	1	11	3	4	5
MPV	9.19	7.01	0.11	141.1	7.63	8.7	9.8
PCT	0.17	0.08	0	0.6	0.1	0.17	0.2
PDW	33.04	9.69	0.16	50	33	37	39.2
Platelet Count	215.69	95.25	20	954	156.3	201	255
Polymorphs	70.48	10.34	17	95	64	71	77
RBC Count	3.77	0.78	2	5.8	3.2	3.7	4.3
RDW SD	21.81	13.04	11.7	68.3	14	15.7	20
WBC Count	7.80	2.92	3	21.6	5.7	7.2	9.1
Eosinophils	2.30	1.14	0	9	2	2	3
Creatinine	6.62	4.28	0	18	2	7	9.98
Urea	109.61	55.50	0	285	66.5	110	148
GFR	23.27	31.23	2.82	209.9	4.90	7.21	28.14
Target	0.61	0.49	0	1	0	1	1

Hyperparameter tuning

Hyperparameter tuning was performed to optimize the performance of the ML models and to prevent overfitting. GridSearchCV technique was employed to systematically explore a predefined set of hyperparameter values for each model.

For each classifier, a range of key hyperparameters such as the number of estimators, maximum tree depth, learning rate, regularization parameters, and kernel settings was defined based on prior knowledge and empirical testing. Optimal combination of hyperparameters was selected using cross-validation on the training data.

This tuning process ensures that each model operates under its best configuration, leading to a fair and reliable comparison across all models. Table 4 presents the hyperparameter search spaces explored for each ML model using GridSearchCV with cross-validation.

Model evaluation strategy

Ensure a reliable and unbiased assessment of model performance, a comprehensive evaluation strategy was adopted. The dataset was initially divided into training (80%) and testing (20%) sets using stratified sampling to preserve the class distribution. The training set was used for model development and hyperparameter tuning, while the testing set was used for final performance evaluation.

In addition to the train–test split, k-fold cross-validation was employed to further validate the robustness of the models. Two different fold strategies i.e., 5-fold and 10-fold were applied. For each fold, dataset was partitioned into k subsets. Subsequently, each subset was used once as a validation set while the remaining subsets were used for training.

This combined evaluation approach ensures that the models are assessed across multiple data splits. Thus reducing the risk of overfitting and providing a more generalized estimate of model performance. Cross-validation was also integrated within the hyperparameter tuning process to ensure optimal model configuration.

Performance metrics

Evaluate the performance of the proposed models, multiple evaluation metrics were employed. Accuracy was used to measure the overall correctness of the predictions. In addition, precision and recall were used to assess the model's ability to correctly identify CKD cases and avoid false positives. F1-score was calculated as the harmonic mean of precision and recall to provide a balanced measure.

Moreover, specificity was used to evaluate the model's ability to correctly identify non-CKD cases, and balanced accuracy was considered to account for class distribution. Matthews Correlation Coefficient (MCC) was also included as a robust metric for binary classification performance.

Furthermore, Receiver Operating Characteristic (ROC) curves and the Area Under the Curve (ROC-AUC) were used to assess the discriminative ability of the models. Precision Recall (PR) curves were also analyzed to evaluate performance under class imbalance. Additionally, calibration analysis was performed to examine the reliability of predicted probabilities. These metrics provide a comprehensive evaluation framework, ensuring reliable and clinically meaningful assessment of model performance. Confidence intervals were also computed to quantify the uncertainty associated with model performance

Statistical validation

Reliability and statistical significance of the obtained results, multiple statistical validation techniques were employed. Confidence intervals (95%) were calculated to quantify the uncertainty associated with model performance and to provide a range within which the true performance is expected to lie.

McNemar's test was applied to compare the performance of competing models on paired prediction outcomes. This test evaluates whether the difference in classification performance between two models is statistically significant.

Furthermore, a one-way Analysis of Variance (ANOVA) test and Tukey HSD Post-hoc Test was conducted to assess whether there are statistically

Table 4 Hyperparameter search space for machine learning models using GridSearchCV

Model	Hyperparameter Search Space
Logistic Regression (LR)	C = {0.001, 0.01, 0.1, 1, 10, 15}; Solver = {lbfgs, liblinear}
Random Forest (RF)	n_estimators = {50, 100, 200, 500}; max_depth = {1–10, None}; min_samples_split = {2, 5, 10}
Support Vector Machine (SVM)	Kernel = {linear, rbf}; C = {0.01, 0.1, 1, 10}; Gamma = {0.001, 0.01, 0.1, 1}
Decision Tree (DT)	max_depth = {1–10, None}; Criterion = {gini, entropy}
K-Nearest Neighbors (KNN)	n_neighbors = {3, 5, 7, 9}; metric = {euclidean, manhattan}
Gradient Boosting	n_estimators = {50, 100, 200}; learning_rate = {0.01, 0.1}; max_depth = {3, 5}
AdaBoost	n_estimators = {50, 100, 200}; learning_rate = {0.01, 0.1, 1}
XGBoost	n_estimators = {50, 100, 200}; max_depth = {3, 5, 7}; learning_rate = {0.01, 0.1}
CatBoost	iterations = {100, 200}; depth = {4, 6, 8}; learning_rate = {0.01, 0.1}

significant differences in performance among multiple models. These statistical analyses provide a robust validation framework, ensuring that the observed performance differences are meaningful and not due to random variation.

eXplainable artificial intelligence

For the interpretability of the proposed models, XAI techniques were employed. Specifically, SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) were used to provide both global and local insights into model predictions.

SHAP was utilized to quantify the contribution of each feature to the model's predictions. It provides a global understanding of feature importance as well as local explanations for individual predictions. SHAP analysis enables identification of key clinical variables influencing CKD prediction.

LIME was applied to generate local explanations by approximating the model behavior around individual instances. This approach helps to interpret how specific feature values contribute to a particular prediction, offering case-level insights.

Combination of SHAP and LIME improves model transparency and supports the clinical interpretability of the results. This ensures that the predictions are aligned with domain knowledge and can be trusted in healthcare applications. These techniques provide transparency in model decision-making, which is essential for clinical adoption.

Multi-criteria decision analysis (TOPSIS)

To provide an objective ranking of ML models based on multiple performance criteria, the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) was employed. Unlike traditional evaluation approaches that rely on a single metric, TOPSIS considers multiple criteria simultaneously to determine the best-performing model.

TOPSIS ranks models by measuring their relative distance from an ideal best solution (maximum performance across all criteria) and an ideal worst solution. Models closer to the ideal best and farther from the ideal worst are assigned higher rankings. This multi-criteria evaluation approach ensures a more comprehensive and unbiased comparison of models, supporting robust model selection for CKD prediction.

Implementation environment

All experiments were conducted using Python in a Jupyter Notebook environment.

Implementation utilized widely adopted scientific computing and ML libraries, including NumPy and Pandas for data processing, and Scikit-learn for model development and evaluation. Advanced models were implemented using XGBoost and CatBoost libraries, while model interpretability was achieved using SHAP and LIME. Data visualization was performed using Matplotlib and Seaborn.

Software tools and library versions used in this research are summarized in Table 5, ensuring transparency and reproducibility of the experimental setup.

Table 5 Software and libraries used in the study

Software and Libraries	Version / Description
Programming Language	Python 3.13.9
NumPy	2.3.5 (for numerical computations)
Pandas	2.3.3 (for data manipulation and preprocessing)
SciPy	1.16.3 (for scientific computing and statistical analysis)
Scikit-learn	1.7.2 (for machine learning modeling and evaluation)
Matplotlib	3.10.6 (for data visualization)
Seaborn	0.13.2 (for statistical data visualization)
XGBoost	3.2.0 (for gradient boosting classification)
SHAP	0.51.0 (for model interpretability and explanation)
LIME	0.2.0.1 (for local model explanation)
Random	Built-in Python module (for reproducibility and sampling)

Results and discussion

To guarantee the quality of data, consistency as well as applicability of the features, the dataset was properly subjected to technical validation methods that employed both visual and statistical inspection. Figure 5 provides a heatmap with the pearson correlation coefficients of all the 22 features. Analysis demonstrates that there are good linear correlations among a number of clinical biomarkers, which supports internal consistency and clinical relevance. It can be seen that haemoglobin, hematocrit and RBC count have high positive correlations ($r = 0.88$ and $r = 0.77$ respectively) indicating documented hematological relations. On the same note, urea and creatinine exhibit a positive and strong correlation ($r \approx 0.71$). These trends are in line with conventional clinical risk factors of diagnosing CKD and other data indicate that the ground truth (class label) in the dataset makes biological sense.

To further justify the hardiness and usefulness of the dataset to use in ML and XAI application, a variety of box plots were created and Fig. 6 presents them.

These graphs provide the distribution and dispersion of essential biomarkers by groups of male, female, CKD and Non-CKD. A number of significant clinical trends appear. Values of creatinine and urea are considerably greater and the values of eGFR are considerably smaller than the normal value in patients with CKD reflecting the deterioration of the renal functioning. Furthermore,

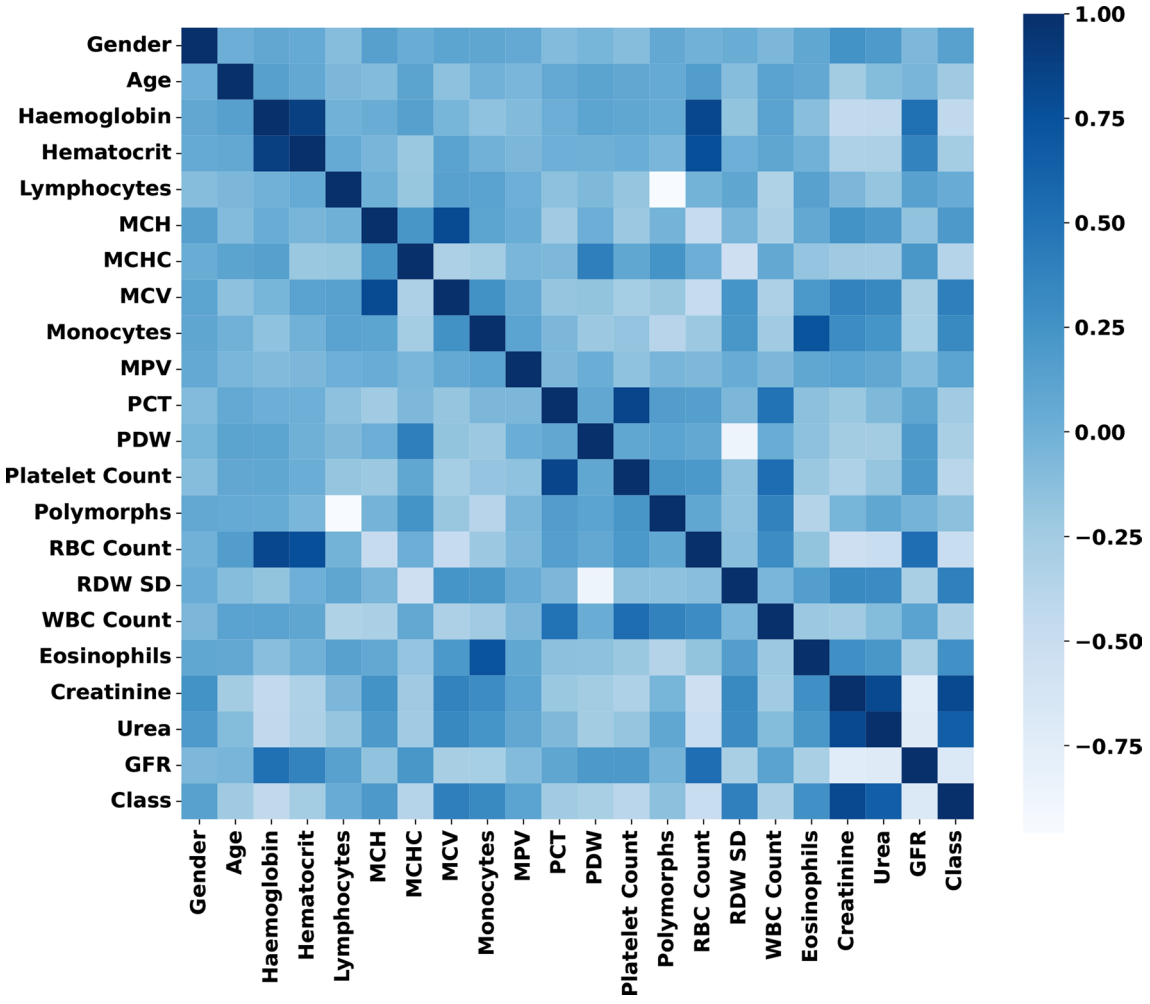


Fig. 5 Visualization of pairwise linear relationships between the dataset's features using pearson correlation coefficients

haemoglobin, hematocrit, and RBC values are reduced in CKD patients particularly in females, which implies that they have CKD-associated anemia. Moreover, there is a significant difference in RDW-SD between the CKD and the control population, but there is also adoption of greater variability in RBC size, which is a widely recognized clinical manifestation of CKD disease.

Box plots, in addition, demonstrate the existence of natural clinical outliers in most variables including creatinine, urea, RDW-SD, and WBC count. These outliers are variability in patient populations in reality and prove that the data is not overly clean. Gender stratification reveals minor but significant differences in biomarker distributions. These variations suggest that gender-based factors may influence how CKD presents in patients. Consequently, gender should be included as a covariate in future modeling, particularly when evaluating an algorithm's fairness or its predictive performance within specific subgroups.

In general, correlation analysis and the box plots also confirm the integrity, completeness, and relevance of the dataset to clinical research and the ML modelling. Likely biological patterns in features, separation ability of classes, and consistency of inter-variable interactions make the dataset suitable to learn robust ML and XAI algorithms without any extra pre-processing and correction. A combination of all these validation observed numbers prove that the dataset has the right clinical variability, internal consistency, and is relevant to the target outcome (CKD classification), which will guarantee its use in further work with MLs and its reliability.

All in all, both the correlation analysis and the boxplots as a whole confirm that the data are valid and complete and can be used in a clinical research study and with ML modeling. The characteristics show predicted biological trends, sensible category segregation, and sensible inter-variable connections, thus the dataset would be useful in training powerful ML and XAI models without further significant preprocessing or correction. All these validation numbers indicate that the dataset exhibits suitable clinical variability, internal consistency, and pertinence to

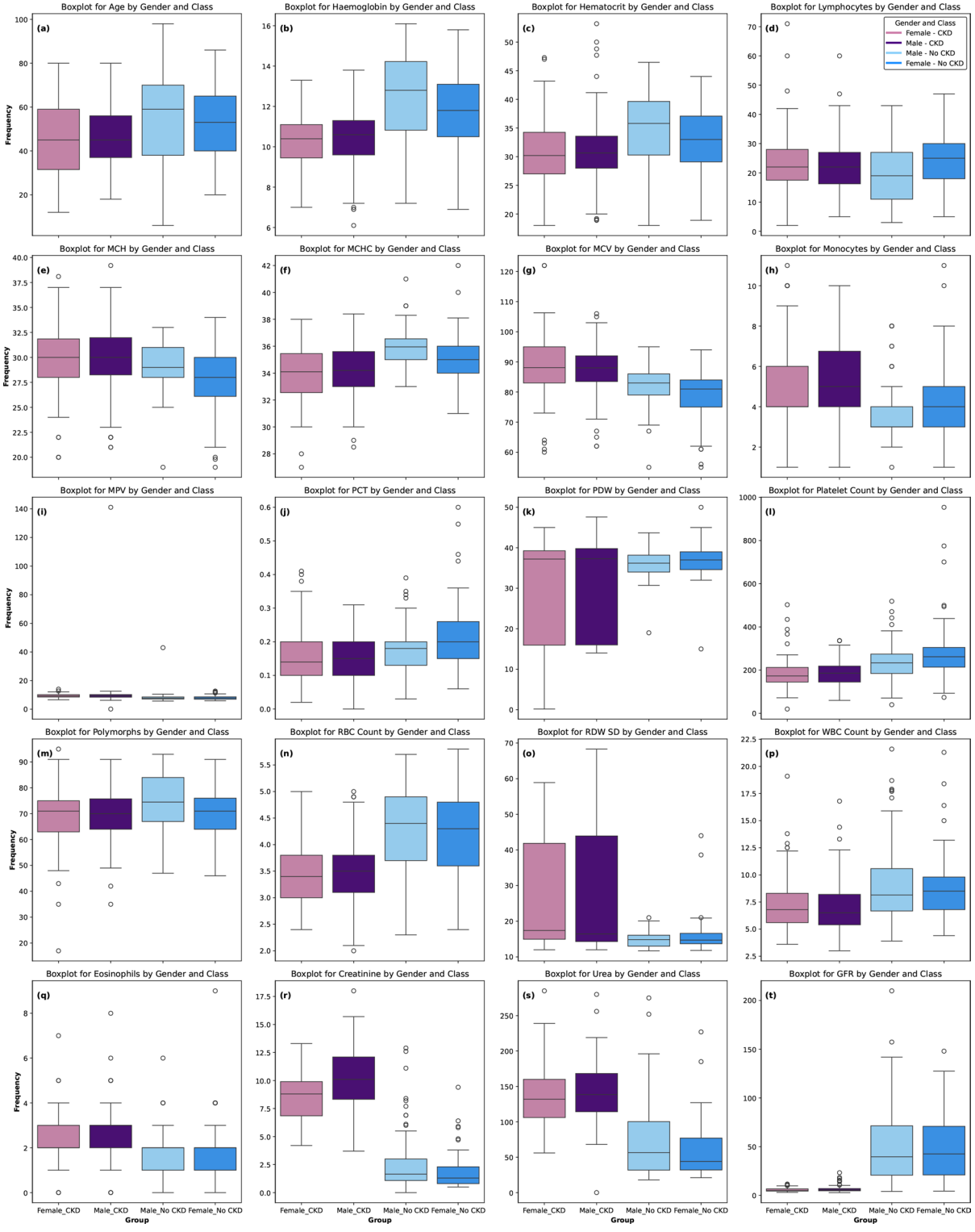


Fig. 6 Boxplots of 20 numerical features stratified by male, female, CKD class, and non-CKD class. Panels (a–t) represent different biomarkers

the intended outcome (CKD classification), which proves its reliability as a downstream device in ML.

The performance of the ML algorithms is analyzed and compared in the following section and conclusions reached on the basis of the findings are discussed in reference to the conventional evaluation metrics. Specific emphasis is also placed on the interpretation of the XAI outputs, where it is also shown which features are of the greatest importance in prediction of CKD.

Feature-wise statistical analysis

To further investigate the statistical significance of individual features in distinguishing CKD and non-CKD patients, both parametric (independent t-test) and non-parametric (Mann–Whitney U test) statistical tests were conducted. The results are summarized in Table 6.

The analysis reveals that several clinical features, including creatinine, glomerular filtration rate (GFR), urea, haemoglobin, and platelet count, show highly significant differences between CKD and non-CKD groups ($p > 0.05$). Notably, creatinine and GFR exhibit extremely low *p*-values, highlighting their strong discriminative power for CKD prediction.

In contrast, features such as lymphocytes and PDW showed non-significant differences, suggesting limited contribution to classification performance. The consistency between t-test and Mann–Whitney test results further strengthens the reliability of these findings.

Table 6 Feature-wise statistical comparison between CKD and non-CKD groups

Feature	CKD Mean	Non-CKD Mean	T-test p-value	Mann-Whitney p-value
GFR	−0.627	0.963	9.61e-82	4.65e-56
Creatinine	0.656	−1.007	4.37e-95	6.91e-56
Urea	0.523	−0.802	1.20e-48	1.12e-40
MPV	0.353	−0.542	5.26e-20	4.06e-21
RBC Count	−0.399	0.612	7.33e-26	4.59e-21
MCV	0.332	−0.510	1.00e-17	5.49e-18
Platelet Count	−0.336	0.516	4.13e-18	8.83e-17
Haemoglobin	−0.351	0.539	8.92e-20	3.39e-16
RDW SD	0.332	−0.510	1.07e-17	2.55e-13
MCHC	−0.301	0.463	1.30e-14	3.77e-13
Monocytes	0.273	−0.418	4.53e-12	6.74e-13
Eosinophils	0.246	−0.377	5.66e-10	1.20e-10
WBC Count	−0.247	0.379	4.58e-10	4.43e-10
Hematocrit	−0.213	0.327	9.05e-08	3.59e-08
PCT	−0.192	0.294	1.68e-06	1.20e-06
Age	−0.184	0.283	4.30e-06	4.85e-06
MCH	0.170	−0.261	2.31e-05	2.72e-05
Polymorphs	−0.105	0.161	0.0094	0.0153
PDW	−0.180	0.276	7.07e-06	0.4188
Lymphocytes	0.022	−0.034	0.5842	0.6920

Overall, this statistical analysis confirms the clinical relevance of key biomarkers and supports their inclusion in the predictive modeling framework.

Assessment of the ML algorithms

Performance of ML models was evaluated using an 80/20 train–test split. Table 7 presents the comparative results of all models using the full feature set across multiple evaluation metrics.

Among all the models, RF achieved the best performance, with an accuracy of 98.75%, precision of 97.96%, recall of 100%, and an F1-score of 98.97%. The model also demonstrated strong balanced accuracy (98.44%) and a high MCC of 0.974, indicating robust and reliable classification performance.

Ensemble models such as Gradient Boosting and CatBoost also performed competitively, achieving high accuracy and F1-scores. However, simpler models such as KNN and SVM showed comparatively lower performance, suggesting that ensemble methods are more effective in capturing complex patterns in clinical data.

Overall, the results indicate that tree-based ensemble models, particularly RF, are highly effective for CKD prediction using structured clinical datasets.

To better understand classification performance, a confusion matrix was created to compare predicted and oracle labels on the test set. It is on this basis that key performance measures such as precision, recall, and F1 score can be computed. The confusion matrix for the RF model used on the experimental data is shown in Fig. 7.

Table 7 presents the experimental results of all ML models (RF, XG boost, CatBoost, Gradient Boosting, LR, AdaBoost, DT, SVM and K-NN) based on the five evaluation parameters of accuracy, precision, recall, F1 score and ROC AUC. All these metrics provide high values with minimal variation in RF which prove that it is robust. CatBoost and XGBoost also perform well, though they are inferior to RF. In contrast, K-NN performs significantly worse according to most metrics, which means that it does not suit the properties of this data set. In general, these findings allow for conclusive finding that RF is the most suitable model in this environment, and comparative graphs and tables can be used to emphasize the relative merits and drawbacks of each algorithm according to several evaluation metrics.

In this investigation, the RF showed performance that was on par with or even better than previous models. RF gives better results because it uses an ensemble learning strategy, which entails constructing several DTs from randomly selected data and attributes. In order to make a final forecast, overfitting is minimized and generalizability is enhanced via averaging or majority voting. As an efficient ML approach, RF excels at handling high-dimensional data, is scalable, and resilient.

Table 7 Performance comparison of machine learning models using 80/20 train–test split (full features)

Model	Accuracy	Precision	Recall	F1-score	Specificity	Bal. Acc	MCC
Random Forest	**0.9875**	**0.9796**	**1.0000**	**0.9897**	**0.9688**	**0.9844**	**0.9742**
Gradient Boosting	0.9750	0.9792	0.9792	0.9792	0.9688	0.9740	0.9479
CatBoost	0.9750	0.9792	0.9792	0.9792	0.9688	0.9740	0.9479
Decision Tree	0.9625	0.9592	0.9792	0.9691	0.9375	0.9583	0.9218
XGBoost	0.9625	0.9787	0.9583	0.9684	0.9688	0.9635	0.9226
Logistic Regression	0.9500	0.9583	0.9583	0.9583	0.9375	0.9479	0.8958
AdaBoost	0.9500	0.9400	0.9792	0.9592	0.9063	0.9427	0.8960
KNN	0.9375	0.9388	0.9583	0.9485	0.9063	0.9323	0.8694
SVM	0.9250	0.9038	0.9792	0.9400	0.8438	0.9115	0.8452

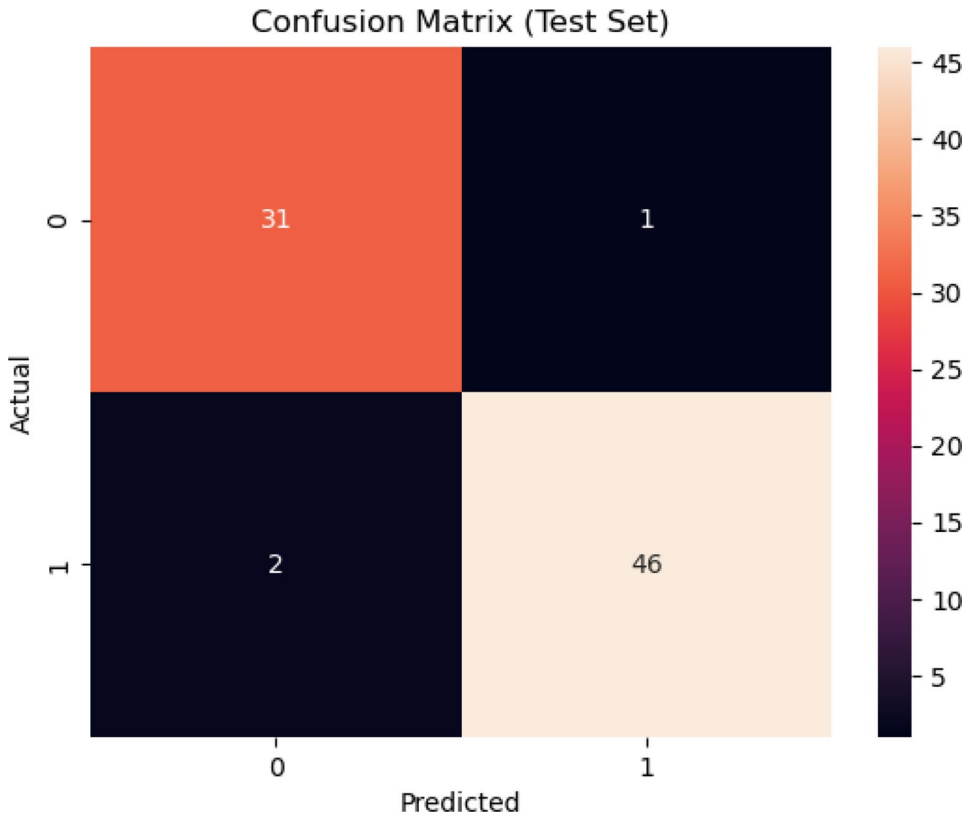


Fig. 7 Confusion matrix of the random forest model on the independent test set, showing high classification performance with minimal misclassifications (1 false positive and 2 false negatives)

To further evaluate the discriminative performance of the proposed model, Receiver Operating Characteristic (ROC) and Precision–Recall (PR) curves were analyzed on the independent test dataset. As shown in Fig. 8, the ROC curve demonstrates excellent classification capability, achieving an area under the curve (AUC) of 0.9863, indicating strong separability between CKD and non-CKD classes. Additionally, the PR curve presented in Figure 9 yields a high PR-AUC of 0.9885, confirming the model's ability to maintain high precision across varying recall levels.

This is particularly important in medical diagnosis, where accurately identifying CKD cases is critical. The high PR-AUC further suggests that the model effectively minimizes false positives while maintaining high sensitivity. Collectively, these results highlight the robustness and reliability of the proposed approach for CKD prediction, especially in clinically sensitive decision-making scenarios.

Feature selection results

Feature selection was performed using three techniques: MI, DT feature importance, and RF feature importance. These methods were applied to identify the most relevant features contributing to CKD prediction. Results from these methods, shown in Table 8 consistently highlighted key clinical variables.

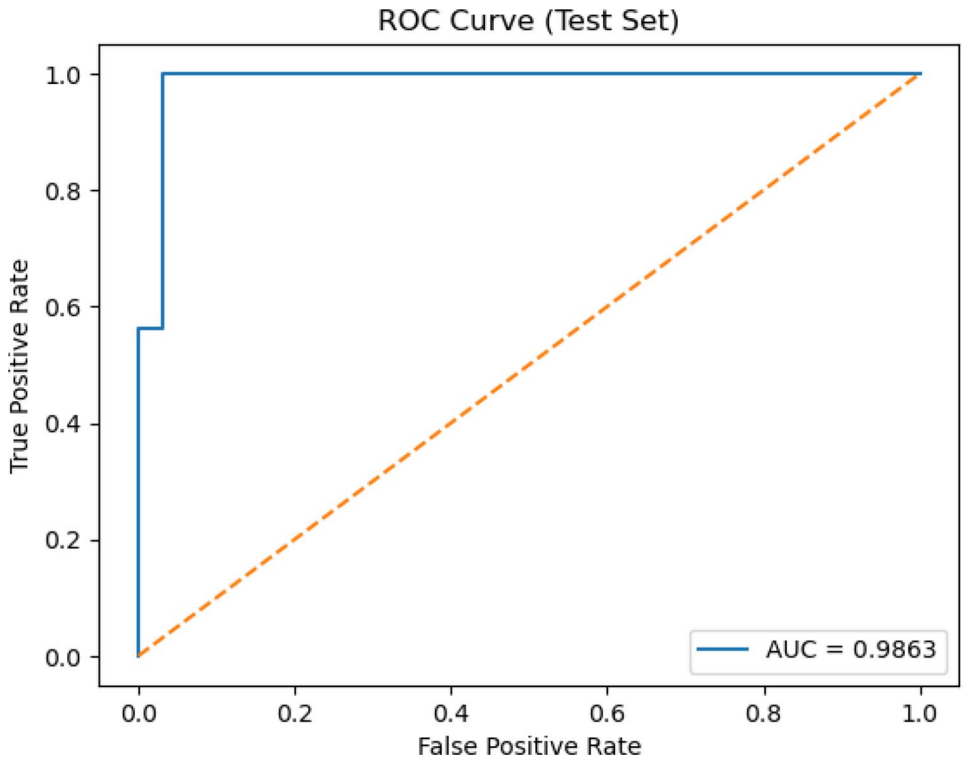


Fig. 8 ROC curve of the random forest model

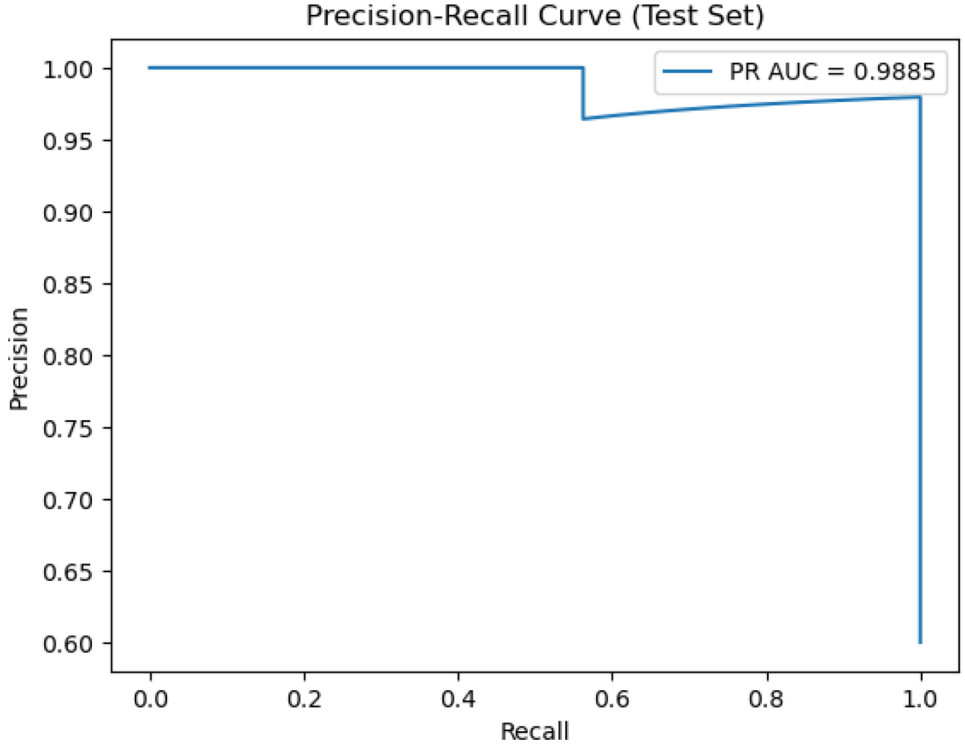


Fig. 9 Precision–recall curve of the random forest model

Table 8 Combined feature importance using Mutual information (MI), Decision Tree (DT), and Random Forest (RF)

Feature	MI Score	DT Importance	RF Importance
GFR	0.4652	0.0183	0.2519
Creatinine	0.4601	0.7605	0.2658
Urea	0.2955	0.0319	0.1047
PDW	0.1789	0.0100	0.0196
Haemoglobin	0.1472	0.0000	0.0557
MPV	0.1327	0.0320	0.0480
RDW SD	0.1319	0.0190	0.0380
RBC Count	0.1029	0.0000	0.0300
Platelet Count	0.0889	0.0537	0.0418
Monocytes	0.0854	0.0000	0.0125
MCV	0.0829	0.0378	0.0255
MCHC	0.0798	0.0000	0.0211
Age	0.0583	0.0088	0.0134
Hematocrit	0.0574	0.0000	0.0153
PCT	0.0366	0.0000	0.0092
Eosinophils	0.0327	0.0066	0.0042
Lymphocytes	0.0175	0.0000	0.0105
WBC Count	0.0104	0.0099	0.0128
Polymorphs	0.0000	0.0000	0.0108
MCH	0.0000	0.0115	0.0080
Gender	0.0000	0.0000	0.0012

Table 9 Comparison of model performance using full features and selected features

Model	Full Features		Selected Features	
	Accuracy	F1-score	Accuracy	F1-score
Random Forest	0.9875	0.9897	0.9875	0.9897
Gradient Boosting	0.9750	0.9792	0.9750	0.9792
XGBoost	0.9625	0.9684	0.9750	0.9792
AdaBoost	0.9500	0.9592	0.9750	0.9796
Logistic Regression	0.9500	0.9583	0.9500	0.9583
KNN	0.9375	0.9485	0.9375	0.9485
SVM	0.9250	0.9400	0.9250	0.9400

Based on the combined analysis, the top features selected for further modeling are GFR, Creatinine, Urea, PDW, Haemoglobin, MPV, RDW SD, RBC Count, Platelet Count and Monocytes. These features are clinically significant and directly associated with kidney function and disease progression. Consistency of feature importance across multiple methods strengthens the reliability of the selected feature subset.

Comparison between full and selected features

Following the identification of the most relevant features (section "Feature selection"), the impact of feature selection on model performance was evaluated by comparing results obtained using the full feature set and the selected subset. Comparative performance of the models under both settings is presented in Table 9.

Results indicate that the performance of the models remains largely consistent even after reducing the number of features. Notably, the RF achieved the same accuracy of 98.75% and F1-score of 98.97% using both full and selected features, demonstrating that the selected features effectively capture the most critical information for CKD prediction.

Similarly, models such as Gradient Boosting, XGBoost, and AdaBoost also maintained comparable or slightly improved performance when using the reduced feature set. This suggests that redundant or less informative features can be removed without negatively affecting model performance.

Overall, feature selection enhances computational efficiency and model interpretability while preserving predictive performance, making it highly suitable for real-world clinical applications.

Cross-validation results

To ensure the robustness and generalizability of the proposed framework, stratified 5-fold and 10-fold cross-validation strategies were employed using the selected top 10 clinical features (Creatinine, GFR, Urea, PDW, Haemoglobin, MPV, RDW SD, Platelet Count, RBC Count, and Monocytes). The results are summarized in Table 10 and Table 11.

Across both validation settings, all models demonstrated consistently high predictive performance, with accuracy values generally exceeding 90% and AUC values approaching 0.98. This confirms that the reduced feature

Table 10 5-fold cross-validation results (Mean ± std) using top 10 features

Model	Acc	Bal Acc	Prec	Rec	Spec	F1	MCC	AUC
Random Forest	0.95 ± 0.02	0.94 ± 0.02	0.95 ± 0.03	0.97 ± 0.03	0.91 ± 0.05	0.96 ± 0.01	0.90 ± 0.03	0.98 ± 0.01
Gradient Boosting	0.93 ± 0.02	0.93 ± 0.02	0.94 ± 0.02	0.95 ± 0.04	0.90 ± 0.03	0.95 ± 0.02	0.86 ± 0.04	0.98 ± 0.01
CatBoost	0.95 ± 0.01	0.94 ± 0.02	0.94 ± 0.03	0.98 ± 0.03	0.90 ± 0.05	0.96 ± 0.01	0.90 ± 0.02	0.99 ± 0.01
Decision Tree	0.91 ± 0.03	0.90 ± 0.03	0.92 ± 0.03	0.93 ± 0.03	0.87 ± 0.06	0.93 ± 0.02	0.81 ± 0.06	0.90 ± 0.03
XGBoost	0.94 ± 0.02	0.93 ± 0.02	0.93 ± 0.03	0.97 ± 0.03	0.89 ± 0.05	0.95 ± 0.01	0.88 ± 0.04	0.99 ± 0.01
Logistic Regression	0.95 ± 0.02	0.94 ± 0.02	0.95 ± 0.03	0.97 ± 0.02	0.92 ± 0.04	0.96 ± 0.01	0.90 ± 0.04	0.98 ± 0.01
AdaBoost	0.95 ± 0.01	0.94 ± 0.01	0.94 ± 0.03	0.97 ± 0.03	0.90 ± 0.05	0.96 ± 0.01	0.89 ± 0.02	0.98 ± 0.01
KNN	0.90 ± 0.04	0.89 ± 0.05	0.89 ± 0.04	0.95 ± 0.02	0.82 ± 0.08	0.92 ± 0.03	0.79 ± 0.09	0.93 ± 0.04
SVM	0.90 ± 0.02	0.88 ± 0.03	0.87 ± 0.02	0.98 ± 0.01	0.78 ± 0.05	0.92 ± 0.02	0.79 ± 0.04	0.95 ± 0.04

Table 11 10-fold cross-validation results (Mean ± std) using top 10 features

Model	Acc	Bal Acc	Prec	Rec	Spec	F1	MCC	AUC
Random Forest	0.95 ± 0.04	0.94 ± 0.05	0.94 ± 0.06	0.98 ± 0.04	0.90 ± 0.11	0.96 ± 0.03	0.89 ± 0.08	0.99 ± 0.01
Gradient Boosting	0.92 ± 0.04	0.92 ± 0.04	0.93 ± 0.04	0.95 ± 0.07	0.89 ± 0.07	0.94 ± 0.04	0.85 ± 0.08	0.98 ± 0.02
CatBoost	0.95 ± 0.05	0.94 ± 0.06	0.94 ± 0.07	0.98 ± 0.05	0.90 ± 0.13	0.96 ± 0.04	0.89 ± 0.09	0.99 ± 0.02
Decision Tree	0.92 ± 0.05	0.92 ± 0.05	0.95 ± 0.06	0.93 ± 0.07	0.91 ± 0.09	0.93 ± 0.04	0.84 ± 0.10	0.92 ± 0.05
XGBoost	0.94 ± 0.05	0.94 ± 0.05	0.94 ± 0.06	0.97 ± 0.05	0.90 ± 0.11	0.95 ± 0.04	0.89 ± 0.09	0.99 ± 0.01
Logistic Regression	0.94 ± 0.03	0.93 ± 0.04	0.95 ± 0.06	0.96 ± 0.05	0.91 ± 0.11	0.95 ± 0.03	0.88 ± 0.07	0.99 ± 0.01
AdaBoost	0.94 ± 0.04	0.94 ± 0.05	0.94 ± 0.06	0.97 ± 0.05	0.90 ± 0.11	0.95 ± 0.03	0.89 ± 0.08	0.99 ± 0.01
KNN	0.91 ± 0.07	0.90 ± 0.08	0.91 ± 0.07	0.95 ± 0.04	0.84 ± 0.13	0.93 ± 0.05	0.81 ± 0.14	0.93 ± 0.07
SVM	0.90 ± 0.05	0.88 ± 0.06	0.88 ± 0.07	0.97 ± 0.04	0.79 ± 0.13	0.92 ± 0.04	0.80 ± 0.10	0.95 ± 0.05

Table 12 Statistical validation results of model performance

Test	Metric	Value
Confidence Interval	Accuracy (RF)	0.9875
	95% CI	[0.9625, 1.0000]
McNemar's Test	Chi-square statistic	1.00
	p-value	1.00
ANOVA Test	F-statistic	2.3487
	p-value	0.1111

set retains strong discriminative capability for CKD prediction.

Across both 5-fold and 10-fold cross-validation settings, CatBoost consistently demonstrated the most balanced and robust performance, achieving the highest MCC and competitive AUC values. Random Forest also showed strong stability and comparable accuracy, while XGBoost achieved the highest AUC in the 10-fold setting, indicating superior ranking capability. Logistic Regression remained highly competitive despite its simplicity, further confirming the strong discriminative power of the selected features. The consistency of results across different cross-validation strategies highlights the reliability and generalizability of the proposed framework.

Statistical validation results

To ensure the robustness and reliability of the proposed framework, a comprehensive statistical validation was performed, as summarized in Table 12 and the Tukey HSD multiple comparison results in 13. The model achieved a high accuracy of 0.9875, with a narrow 95% confidence interval of [0.9625, 1.0000], indicating strong and consistent predictive performance across resampled datasets. Furthermore, McNemar's test yielded a chi-square statistic of 1.00 with a p-value of 1.00, demonstrating that there is no statistically significant difference between the predictions of the compared models on the test set, and highlighting a high level of agreement in classification outcomes.

In addition, a one-way ANOVA test was conducted to compare model performance across cross-validation folds. The results ($F = 2.3487$, $p = 0.111$) indicate that there is no statistically significant difference among the evaluated models. This finding suggests that all models exhibit comparable predictive capabilities when applied to the given dataset. To further investigate pairwise differences, a Tukey HSD post-hoc analysis was performed, the results of which are presented in the corresponding table. Notably, all pairwise comparisons resulted in non-significant outcomes (reject = False), with overlapping confidence intervals across all model combinations.

Collectively, these statistical findings indicate that the evaluated ML models demonstrate consistent and stable performance, with minimal variation across different algorithms. The absence of statistically significant differences does not imply a lack of model effectiveness; rather, it reflects the high predictive strength of the dataset and the strong agreement among classifiers. In such scenarios, model selection can be guided by secondary criteria such as interpretability, computational efficiency, and clinical applicability. Based on overall performance metrics and consistency, the selected model is considered a reliable and competitive approach for CKD prediction.

The absence of statistically significant differences among models can be attributed to the strong predictive signal present in the dataset, coupled with the relatively small sample size and high agreement among model predictions. These factors limit the sensitivity of statistical tests such as McNemar's test and ANOVA. Nevertheless, the consistent performance across multiple models indicates the robustness and reliability of the proposed feature set and predictive framework.

The statistical analysis of clinical features, presented in Table 14, reveals that several biomarkers exhibit highly significant differences between CKD and non-CKD groups. Notably, features such as Creatinine ($d = -2.85$), GFR ($d = 1.90$), and Urea ($d = -1.72$) demonstrate exceptionally large effect sizes, indicating strong discriminative capability. These findings are consistent with established clinical knowledge, where elevated creatinine and urea levels, along with reduced GFR, are key indicators of impaired kidney function.

Furthermore, additional hematological parameters including RBC Count, Haemoglobin, Platelet Count, and RDW SD also show large to moderate effect sizes,

Table 13 Tukey HSD Post-hoc test results for model comparison (accuracy scores)

Model 1	Model 2	Mean Diff	p-adj	Lower	Upper	Reject
AdaBoost	CatBoost	0.0000	1.0000	−0.0747	0.0747	False
AdaBoost	Decision Tree	−0.0472	0.5400	−0.1219	0.0276	False
AdaBoost	Gradient Boosting	−0.0064	1.0000	−0.0811	0.0684	False
AdaBoost	KNN	−0.0534	0.3675	−0.1282	0.0213	False
AdaBoost	Logistic Regression	0.0031	1.0000	−0.0716	0.0779	False
AdaBoost	Random Forest	−0.0001	1.0000	−0.0748	0.0746	False
AdaBoost	SVM	−0.0375	0.8026	−0.1122	0.0372	False
AdaBoost	XGBoost	0.0031	1.0000	−0.0716	0.0779	False
CatBoost	Decision Tree	−0.0472	0.5400	−0.1219	0.0276	False
CatBoost	Gradient Boosting	−0.0064	1.0000	−0.0811	0.0684	False
CatBoost	KNN	−0.0534	0.3675	−0.1282	0.0213	False
CatBoost	Logistic Regression	0.0031	1.0000	−0.0716	0.0779	False
CatBoost	Random Forest	−0.0001	1.0000	−0.0748	0.0746	False
CatBoost	SVM	−0.0375	0.8026	−0.1122	0.0372	False
CatBoost	XGBoost	0.0031	1.0000	−0.0716	0.0779	False
Decision Tree	Gradient Boosting	0.0408	0.7197	−0.0339	0.1156	False
Decision Tree	KNN	−0.0062	1.0000	−0.0810	0.0685	False
Decision Tree	Logistic Regression	0.0503	0.4511	−0.0244	0.1250	False
Decision Tree	Random Forest	0.0471	0.5429	−0.0277	0.1218	False
Decision Tree	SVM	0.0097	1.0000	−0.0651	0.0844	False
Decision Tree	XGBoost	0.0503	0.4511	−0.0244	0.1250	False
Gradient Boosting	KNN	−0.0471	0.5429	−0.1218	0.0277	False
Gradient Boosting	Logistic Regression	0.0095	1.0000	−0.0653	0.0842	False
Gradient Boosting	Random Forest	0.0062	1.0000	−0.0685	0.0810	False
Gradient Boosting	SVM	−0.0311	0.9198	−0.1059	0.0436	False
Gradient Boosting	XGBoost	0.0095	1.0000	−0.0653	0.0842	False
KNN	Logistic Regression	0.0566	0.2922	−0.0182	0.1313	False
KNN	Random Forest	0.0533	0.3700	−0.0214	0.1281	False
KNN	SVM	0.0159	0.9989	−0.0588	0.0907	False
KNN	XGBoost	0.0566	0.2922	−0.0182	0.1313	False
Logistic Regression	Random Forest	−0.0032	1.0000	−0.0780	0.0715	False
Logistic Regression	SVM	−0.0406	0.7250	−0.1154	0.0341	False
Logistic Regression	XGBoost	0.0000	1.0000	−0.0747	0.0747	False
Random Forest	SVM	−0.0374	0.8049	−0.1121	0.0373	False
Random Forest	XGBoost	0.0032	1.0000	−0.0715	0.0780	False
SVM	XGBoost	0.0406	0.7250	−0.0341	0.1154	False

reinforcing their relevance in distinguishing CKD patients from healthy individuals. The narrow confidence intervals associated with these features indicate low variability and high stability of the dataset.

Importantly, the presence of multiple features with large effect sizes provides a clear explanation for the high and consistent performance observed across all ML models. Since the underlying data exhibits strong class separability, different classifiers are able to achieve similarly high predictive accuracy. This, in turn, explains the absence of statistically significant differences in model comparison tests such as McNemar's test, ANOVA, and Tukey HSD.

Therefore, the lack of statistical significance among models should not be interpreted as a limitation; rather, it reflects the strength and robustness of the selected feature set. The results indicate that the predictive performance is primarily driven by the inherent quality and discriminative power of the clinical features, rather than dependence on a specific ML algorithm. This enhances the reliability and generalizability of the proposed framework for CKD prediction.

Multi-criteria model ranking (TOPSIS)

To provide a comprehensive and objective comparison of ML models, the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) was employed. Unlike traditional evaluation approaches that rely on a single performance metric, TOPSIS considers multiple evaluation criteria simultaneously to rank models.

In this study, key performance metrics, including accuracy, precision, recall, F1-score, balanced accuracy,

Table 14 Statistical comparison of clinical features between CKD and non-CKD groups

Feature	CKD (Mean ± CI)	Non-CKD (Mean ± CI)	Effect Size (d)
Creatinine	9.42 (9.09–9.76)	2.31 (1.95–2.66)	−2.85
GFR	6.10 (5.77–6.44)	49.63 (43.89–55.37)	1.90
Urea	138.52 (133.50–143.55)	65.22 (57.87–72.57)	−1.72
RBC Count	3.46 (3.38–3.53)	4.24 (4.12–4.37)	1.16
Haemoglobin	10.41 (10.23–10.59)	12.08 (11.75–12.40)	0.99
Platelet Count	185.88 (177.88–193.87)	261.45 (243.08–279.81)	0.86
RDW SD	26.03 (24.11–27.94)	15.34 (14.76–15.92)	−0.89
MCV	87.93 (86.80–89.06)	80.36 (79.16–81.57)	−0.90
Monocytes	5.14 (4.88–5.40)	3.78 (3.52–4.05)	−0.70
WBC Count	7.08 (6.78–7.38)	8.89 (8.37–9.42)	0.65
PDW	30.76 (29.30–32.23)	36.54 (35.94–37.14)	0.62
PCT	0.15 (0.15–0.16)	0.19 (0.18–0.21)	0.50
Age	46.29 (44.37–48.21)	54.03 (51.24–56.81)	0.48
MCH	29.78 (29.38–30.18)	28.51 (28.06–28.96)	−0.42
Eosinophils	2.55 (2.41–2.68)	1.92 (1.73–2.10)	−0.57
Polymorphs	69.36 (68.06–70.66)	72.20 (70.57–73.84)	0.28
MPV	9.86 (8.76–10.95)	8.17 (7.68–8.65)	−0.24
Lymphocytes	22.92 (21.73–24.11)	22.20 (20.67–23.72)	−0.08

Table 15 TOPSIS-based ranking of machine learning models

Model	TOPSIS Score	Rank
Random Forest	**1.0000**	**1**
Gradient Boosting	0.7977	2
CatBoost	0.7977	2
XGBoost	0.6268	4
Decision Tree	0.6267	5
Logistic Regression	0.4657	6
AdaBoost	0.4306	7
KNN	0.2816	8
SVM	0.1315	9

specificity, and MCC, were used to construct the decision matrix. Each model was evaluated according to these criteria to determine its relative proximity to the ideal best and ideal worst solutions.

Ranking results obtained using TOPSIS are presented in Table 15. RF model achieved the highest TOPSIS score, indicating its superior overall performance across all evaluation metrics. Gradient Boosting and CatBoost ranked second, while XGBoost and DT showed moderate performance.

These results demonstrate that RF is the most reliable and well-balanced model for CKD prediction when multiple performance criteria are considered simultaneously. To provide a comprehensive and objective comparison of ML models, the Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS) was employed. Unlike traditional evaluation approaches that rely on a single performance metric, TOPSIS considers multiple evaluation criteria simultaneously to rank models.

Assessment of the XAI approach

The RF has the ability to revolutionize healthcare in the face of rapidly growing medical data and patient demands for more personalized treatment. The blazing-fast execution and exceptional prediction performance set it apart from other ML choices. Figure 10 presents the SHAP summary plot, which highlights the feature importance ranking for the RF.

On one side of the graph we can see the SHAP values, and on the other side we can see the feature rankings. The greater the SHAP value of a particular feature the larger its effect on the results of the model. Whereas, the fewer its contribution the closer it is to the bottom of the plot.

Figure 11 shows the SHAP beeswarm plot, provides both a global view and the effect of individual feature, accordinlgy, the relative contribution of each variable to be seen. Moreover, this plot demonstrates how various feature-values change the prediction of the model. This is useful in not only explaining the variables with the greatest role, but also the interaction between their values resulting in CKD classification.

Figure 12 shows SHAP waterfall plot of CKD prediction in which features are ranked by their impact on prediction thorough SHAP values. On the whole, the features have an overall positive impact on the ultimate probability of CKD and eGFR stands out as the most influential factor.

One can quantify the contributions of each feature to the behavior of the model in the dataset level by summing up the SHAP values of all instances. Parallelly, the application of LIME was used to provide the prediction of an individual patient as shown in Fig. 13. The contributing features are displayed in this example, showing

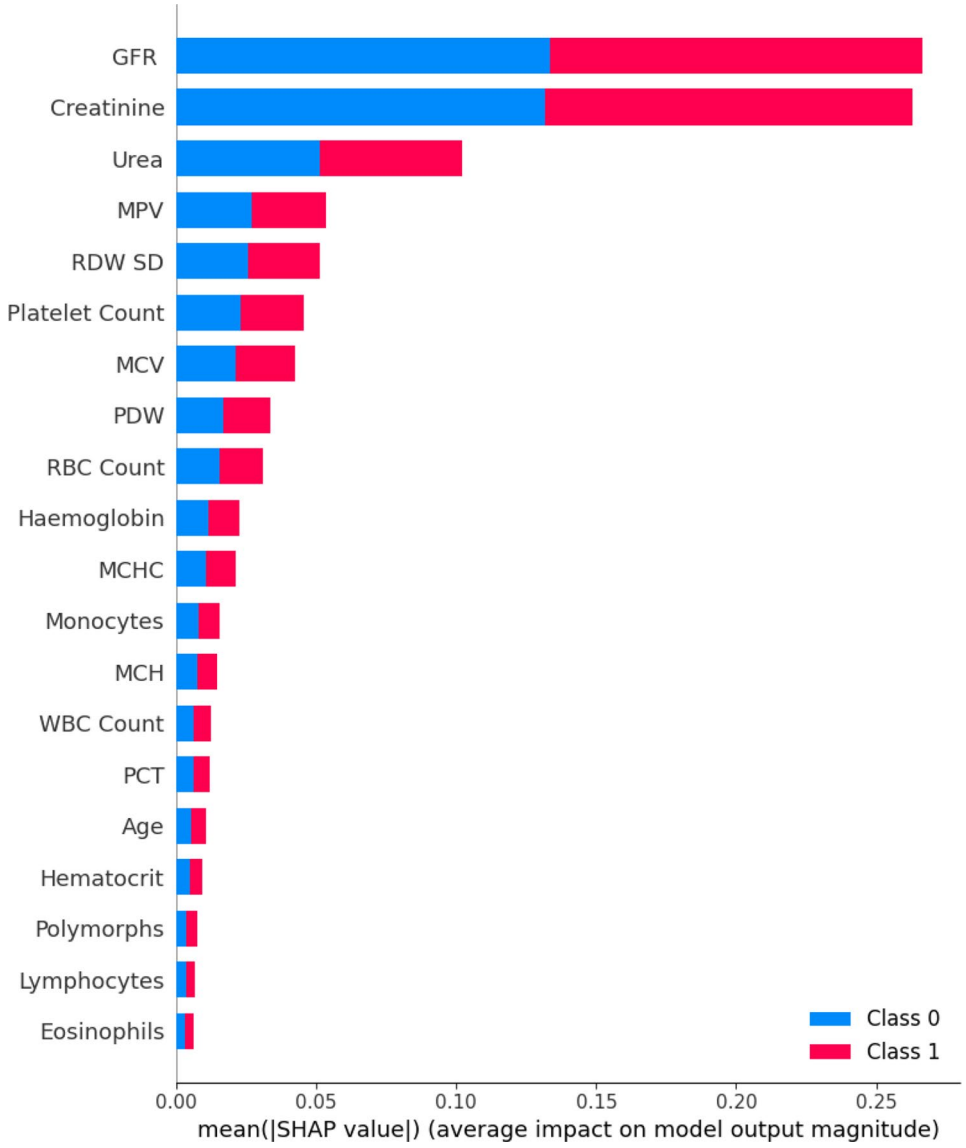


Fig. 10 SHAP-based explainability for CKD prediction

their actual values as to whether they incline the prediction to CKD (orange) or non-CKD (blue), thus making the decision to the particular case more understandable.

This research has built and tested eight ML models based on a dataset that consisted of twenty-two clinical variables, where patients were classified as having CKD and non-CKD. The RF was the most successful of all the models. In order to allay the most frequently stated apprehensions about the lack of transparency regarding the ML models, we incorporated the explainability algorithms, namely SHAP and LIME, into the RF model. These methods allowed improving the interpretability of the CKD predictions, including the fact that they offered details, patient-level and global information about how specific clinical parameters influence the model decisions as well as positioning and the relative significance of the key elements of the profile.

Results of this study demonstrate that ML models, particularly ensemble methods, are highly effective for the early prediction of CKD. Among all evaluated models, RF consistently achieved the best performance across multiple evaluation strategies, including train–test split, cross-validation, and multi-criteria ranking using TOPSIS. This superior performance can be attributed to its ability to capture complex non-linear relationships and interactions among clinical features.

The feature selection analysis revealed that key clinical indicators such as creatinine, GFR, and urea play a significant role in CKD prediction. These findings are consistent with established medical knowledge, as these

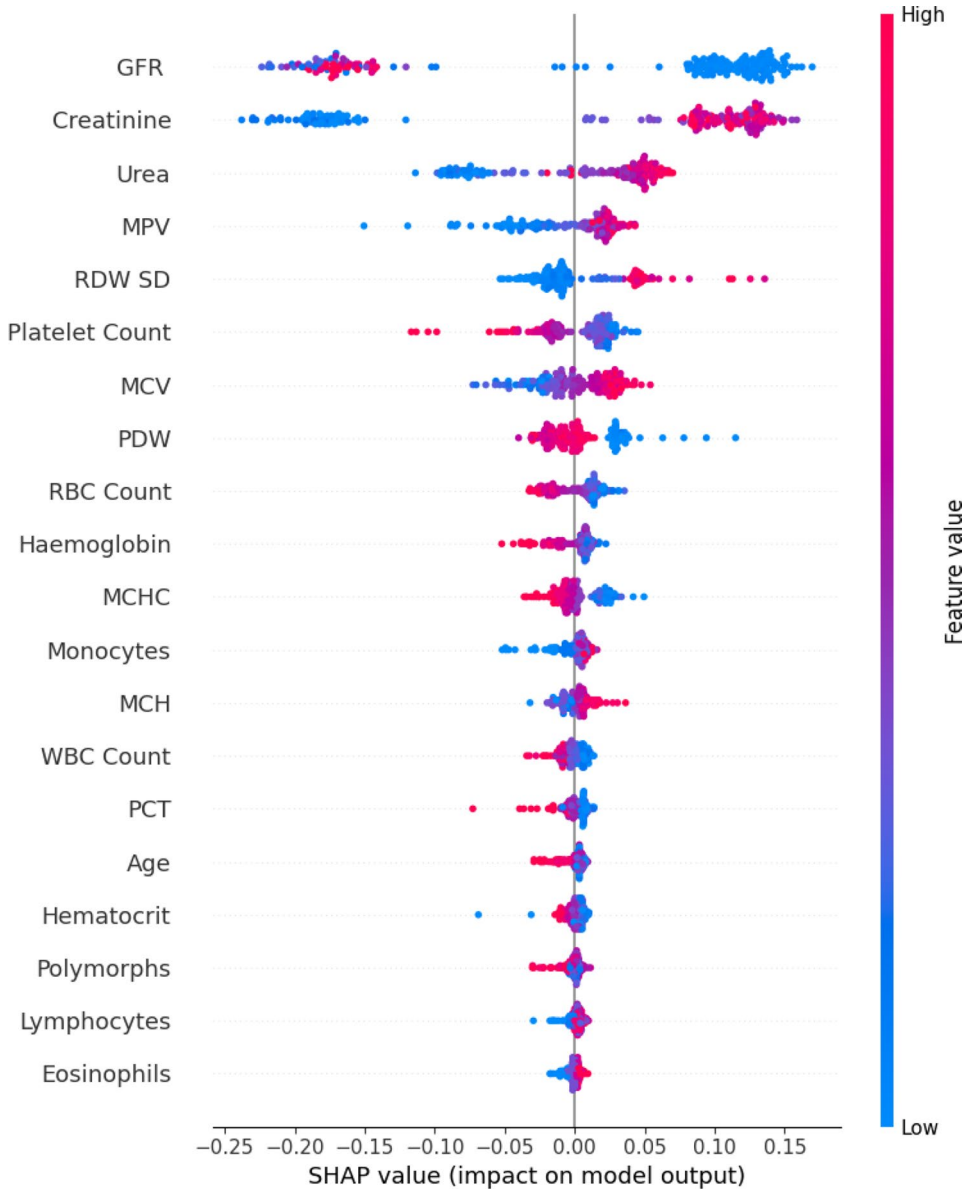


Fig. 11 Beeswarm plot showing the effect of input variables on prediction performance

biomarkers are directly associated with kidney function and disease progression.

Furthermore, the results demonstrate that reducing the feature set does not significantly affect model performance, indicating that a compact set of clinically relevant features is sufficient for accurate prediction. This has important implications for real-world deployment, where reducing the number of required tests can lower costs and improve efficiency.

XAI analysis using SHAP and LIME further supports the reliability of the model by providing interpretable insights into feature contributions. The alignment of these explanations with clinical understanding enhances trust in the model's predictions and supports its potential use in clinical decision support systems.

Overall, the proposed framework provides a reliable, interpretable, and efficient approach for CKD prediction, with strong potential for practical implementation in healthcare settings.

Conclusion

This study presents a comprehensive and XAI-ML framework for the early prediction of CKD using clinical data. A wide range of ML models were evaluated using multiple validation strategies, including train–test split and k-fold cross-validation, along with rigorous statistical validation techniques.

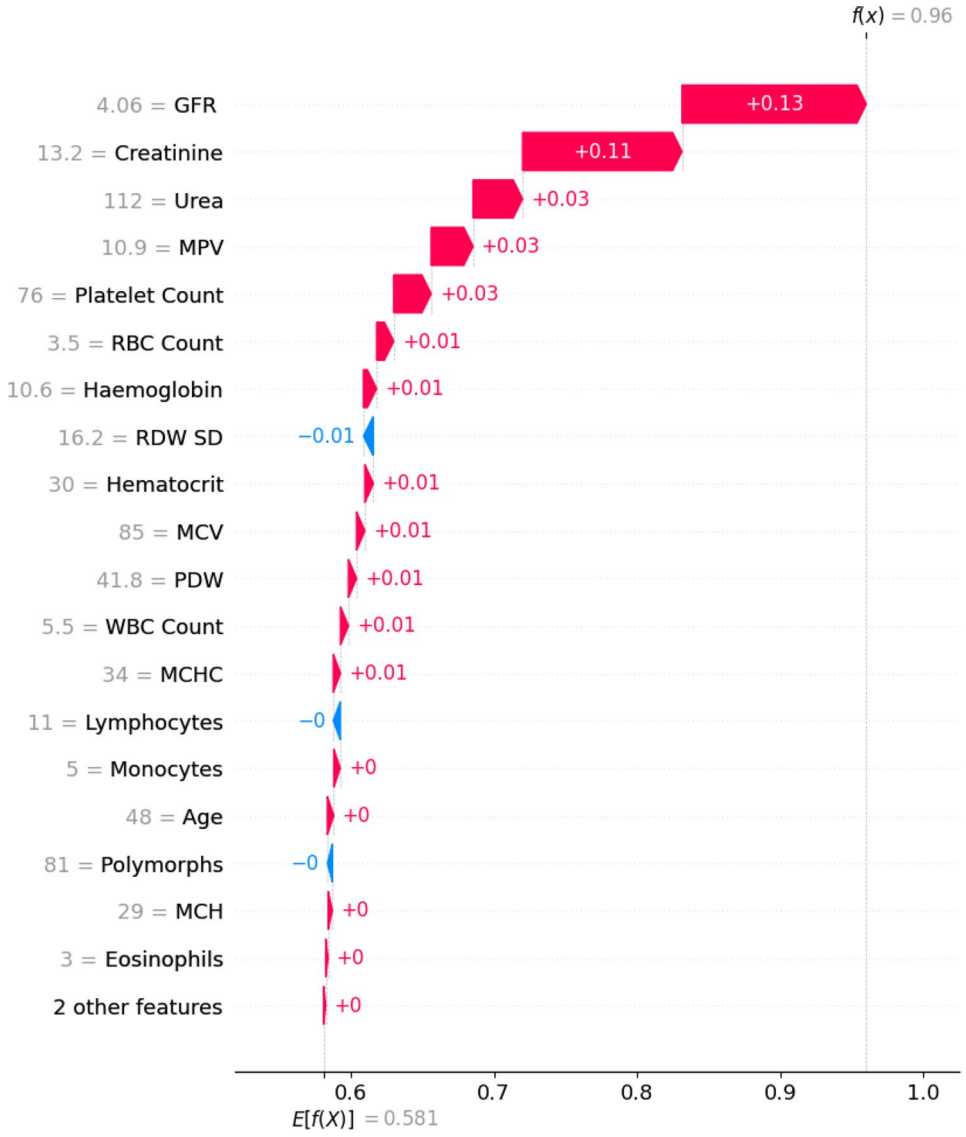


Fig. 12 SHAP waterfall plot illustrating feature contributions for a single prediction

The results demonstrate that ensemble methods, particularly RF, achieve superior performance across multiple evaluation metrics. The application of feature selection techniques further showed that a reduced set of clinically relevant features such as creatinine, GFR, and urea is sufficient for accurate CKD prediction, improving efficiency without compromising performance.

The integration of XAI methods, including SHAP and LIME, provided both global and local interpretability, enhancing the transparency and clinical relevance of the model. Additionally, the use of the TOPSIS method enabled objective model ranking based on multiple criteria, offering a robust approach for model selection.

Overall, the proposed framework demonstrates strong predictive performance, reliability, and interpretability, making it a promising tool for clinical decision support.

Limitations and future work

Although preprocessing and validation to guarantee quality of data is done carefully and by expert individuals, there are still a few limitations. First, the data was collected in one healthcare center (PKC), and this could limit the external validity of the results to other groups or care environments. Second, it has a relatively small sample size (398 records) that might not allow the training of more complex models of ML, due to overfitting. What is more, in spite of the confirmed diagnosis performed by medical workers, the clinical judgment might be different across institutions as well as contain some minor biases. The dataset also does not mean any information of potentially significant variables like medications, lifestyle behavior, and comorbid conditions all of which may contribute to the development and progression of CKD.

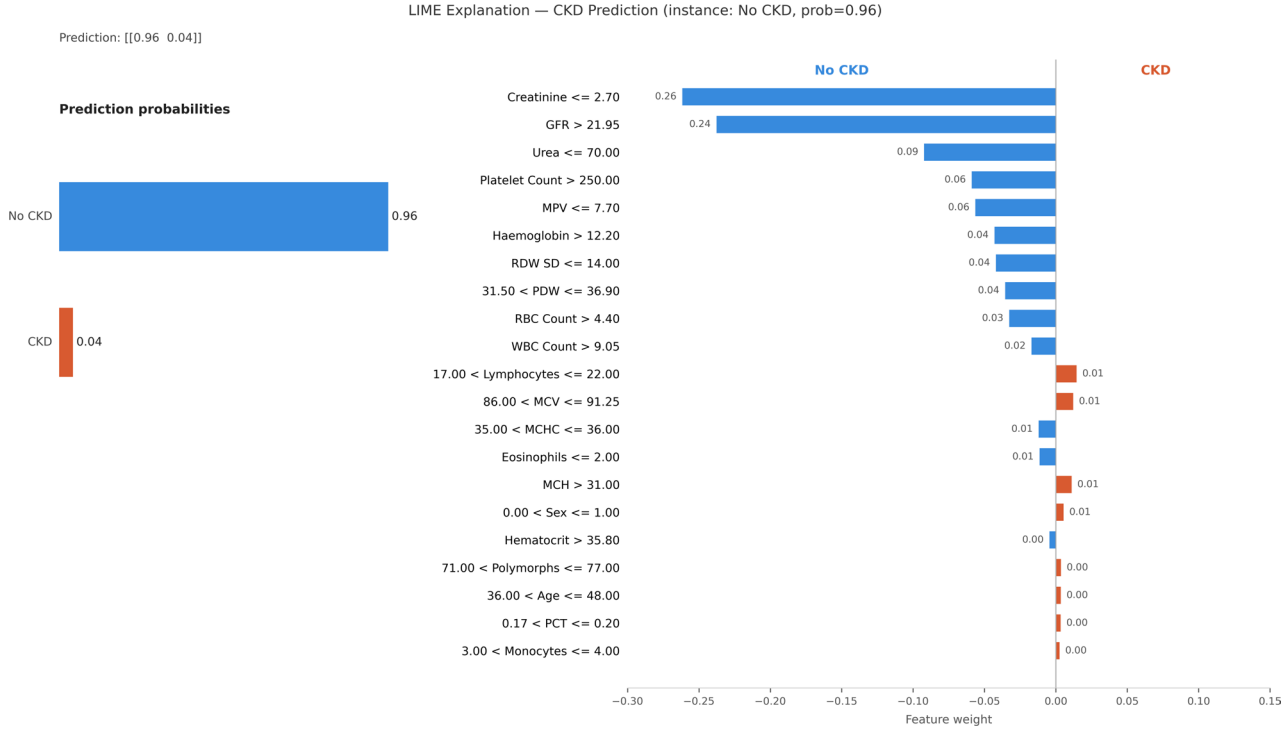


Fig. 13 LIME scores highlighting locally important features

The further research needs to be directed at the development of the dataset of several centers and different populations and extend the spectrum of clinical, treatment, and lifestyle factors. Such methods would enable to enhance the resiliency, external validity and clinical applicability of the.

Supplementary information

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Supplementary Material 1

Authors contribution

Conceptualization, M.R., N.M., and R.N.; Data curation, M.R., R.N., M.A.K. and M.F.; Formal analysis, M.R., R.N., M.A.K, and N.M.; Funding acquisition, N.M.; Investigation, M.R., R.N., M.F., N.M. and A.A; Methodology, M.R., R.N., A.A. and N.M.; Project administration, M.R., R.N., N.M. and A.A; Software, M.R., M.A.K., A.A., and N.M.; Supervision, R.N., A.A. and N.M.; Validation, M.R., M.A.K., M.F., A.A., N.M. and R.N.; Visualization, M.R., M.A.K., M.F., N.M., and R.N.; Writing-original draft, M.R., M.A.K., R.N. and N.M.; Writing-review and editing, R.N., A.A. and N.M.

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Data availability

This manuscript utilizes dataset, which is available at: https://doi.org/10.6084/m9.figshare.29603663, code is available at GitHub repository https://github.com/rizwanawan612/CKD-Classification-ML-XAI and The developed web-based application for CKD prediction is available at: https://huggingface.co/spaces/mcf1801083/CKD_Prediction_System.

Declarations

Consent for publication

Not applicable because no identifiable patient information is included.

Competing interests

The authors declare no competing interests.

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References

1. Ma X, Liu R, Xi X, Zhuo H, Gu Y. Global burden of chronic kidney disease due to diabetes mellitus, 1990-2021, and projections to 2050. Front Endocrinol. 2025;16:16–2025. https://doi.org/10.3389/fendo.2025.1513008.
2. Nature. Kidney disease is a worldwide killer. treat it that way. Nature. 2025;640(8060):858. https://doi.org/10.1038/d41586-025-01237-2.
3. Foreman KJ, Marquez N, Dolgert A, Fukutaki K, Fullman N, McGaughey M, et al. Forecasting life expectancy, years of life lost, and all-cause and cause-specific mortality for 250 causes of death: reference and alternative scenarios for 2016–40 for 195 countries and territories. Lancet. 2018;392(10159):2052–90. https://doi.org/10.1016/S0140-6736(18)31694-5.
4. Stevens PE, Ahmed SB, Carrero JJ, Foster B, Francis A, Hall RK, et al. Kdigo 2024 clinical practice guideline for the evaluation and management of chronic

kidney disease. Kidney Int. 2024;105(4):S117–314. https://doi.org/10.1016/j.kint.2023.10.018.
5. Bello AK, Okpechi IG, Levin A, Ye F, Damster S, Arruebo S, et al. An update on the global disparities in kidney disease burden and care across world countries and regions. Lancet Global Health. 2024;12(3):e382–95. https://doi.org/10.1016/S2214-109X(23)00570-3.
6. Kovesdy CP. Epidemiology of chronic kidney disease: an update 2022. Kidney Int Suppl. 2022;12(1):7–11. https://doi.org/10.1016/j.kisu.2021.11.003.
7. Rady E-HA, Anwar AS. Prediction of kidney disease stages using data mining algorithms. Inf Med Unlocked. 2019;15:100178. https://doi.org/10.1016/j.imu.2019.100178.
8. Caravaca-Fontan F, Azevedo L, Luna E, Caravaca F. Patterns of progression of chronic kidney disease at later stages. Clin Kidney J. 2018;11(2):246–53. https://doi.org/10.1093/ckj/sfx083.
9. Eckardt K-U, Coresh J, Devuyst O, Johnson RJ, Kottgen A, Levey AS, et al. Evolving importance of kidney disease: from subspecialty to global health burden. Lancet. 2013;382(9887):158–69. https://doi.org/10.1016/S0140-6736(13)60439-0.
10. Akter S, Habib A, Islam MA, Hossen MS, Fahim WA, Sarkar PR, et al. Comprehensive performance assessment of deep learning models in early prediction and risk identification of chronic kidney disease. IEEE Access. 2021;9:165184–206. https://doi.org/10.1109/ACCESS.2021.3129491.
11. Heung M, Chawla LS. Predicting progression to chronic kidney disease after recovery from acute kidney injury. Curr Opin Nephrol Hypertens. 2012;21(6):628–34. https://doi.org/10.1097/MNH.0b013e3283588f24.
12. Cueto-Manzano AM, Cortes-Sanabria L, Martinez-Ramirez HR, Rojas- Campos E, Gomez-Navarro B, Castillero-Manzano M. Prevalence of chronic kidney disease in an adult population. Arch Med Res. 2014;45(6):507–13. https://doi.org/10.1016/j.arcmed.2014.06.007.
13. Hsu RK, Powe NR. Recent trends in the prevalence of chronic kidney disease: not the same old song. Curr Opin Nephrol Hypertens. 2017;26(3):187–96. https://doi.org/10.1097/MNH.0000000000000315.
14. Chen Z, Zhang X, Zhang Z. Clinical risk assessment of patients with chronic kidney disease by using clinical data and multivariate models. Int Urol Nephrol. 2016;48(12):2069–75. https://doi.org/10.1007/s11255-016-1346-4.
15. Naqvi AA, Hassali MA, Aftab MT. Epidemiology of rheumatoid arthritis, clinical aspects and socio-economic determinants in Pakistani patients: a systematic review and meta-analysis. J Pak Med Assoc. 2019;69(3):389–98.
16. Mallamaci F, Tripepi G. Risk factors of chronic kidney disease progression: between old and new concepts. JCM. 2024;13(3):678. https://doi.org/10.3390/jcm13030678.
17. Alturki N, Altamimi A, Umer M, Saidani O, Alshardan A, Alsubai S, et al. Improving prediction of chronic kidney disease using knn imputed smote features and trionet model. Cmes-Comp Model Eng. 2024;139(3):3513–34. https://doi.org/10.32604/cmes.2023.045868.
18. Dritsas E, Trigka M. Machine learning techniques for chronic kidney disease risk prediction. BDCC. 2022;6(3):98. https://doi.org/10.3390/bdcc6030098.
19. Yousif SMA, Halawani HT, Amoudi G, Birkea FMO, Almunajam AMR, Elhag AA. Early detection of chronic kidney disease using eurygasters optimization algorithm with ensemble deep learning approach. Alex Eng J. 2024;100:220–31. https://doi.org/10.1016/j.aej.2024.05.011.
20. Pal S. Chronic kidney disease prediction using machine learning techniques. Biomed Mater Devices. 2023;1(1):534–40. https://doi.org/10.1007/s44174-022-00027-y.
21. Shlipak MG, Tummalapalli SL, Boulware LE, Grams ME, Ix JH, Jha V, et al. The case for early identification and intervention of chronic kidney disease: conclusions from a kidney disease: improving global outcomes (kdigo) controversies conference. Kidney Int. 2021;99(1):34–47. https://doi.org/10.1016/j.kint.2020.10.012.
22. Sawhney R, Malik A, Sharma S, Narayan V. A comparative assessment of artificial intelligence models used for early prediction and evaluation of chronic kidney disease. Decis Analytics J. 2023;6:100169. https://doi.org/10.1016/j.dajour.2023.100169.
23. Qaseem A, Hopkins RH, Sweet DE, Starkey M, Shekelle P. Screening, monitoring, and treatment of stage 1 to 3 chronic kidney disease: a clinical practice guideline from the American college of physicians. Ann Intern Med. 2013;159(12):835–47. https://doi.org/10.7326/0003-4819-159-12-201312170-00726.
24. Elkholy SMM, Rezk A, Saleh AAEF. Early prediction of chronic kidney disease using deep belief network. IEEE Access. 2021;9:135542–49. https://doi.org/10.1109/ACCESS.2021.3114306.
25. Silva Selistre L, Rech DL, Souza V, Iwaz J, Lemoine S, Dubourg L. Diagnostic performance of creatinine-based equations for estimating glomerular filtration rate in adults 65 years and older. JAMA Intern Med. 2019;179(6):796–804. https://doi.org/10.1001/jamainternmed.2019.0223.
26. Silveira ACMD, Sobrinho, LDD, Barros Costa E, Pinheiro ME, Perkusich A, et al. Exploring early prediction of chronic kidney disease using machine learning algorithms for small and imbalanced datasets. Appl Sci. 2022;12(7):3673. https://doi.org/10.3390/app12073673.
27. Yadav P, Sharma SC. HFBO-KSELM: hybrid flash butterfly optimization-based kernel softplus extreme learning machine for classification of chronic kidney disease. J Sport Hist Supercomput. 2023;79(15):17146–69. https://doi.org/10.1007/s11227-023-05337-6.
28. Hosseinzadeh M, Koohpayehzadeh J, Bali AO, Asghari P, Souri A, Mazaherinezhad A, et al. A diagnostic prediction model for chronic kidney disease in internet of things platform. Multimed Tools Appl. 2021;80(11):16933–50. https://doi.org/10.1007/s11042-020-09049-4.
29. Halder RK, Uddin MN, Uddin MA, Aryal S, Saha S, Hossen R, et al. Ml-ckdp: machine learning-based chronic kidney disease prediction with smart web application. J Pathol Inf. 2024;15:100371. https://doi.org/10.1016/j.jpi.2024.100371.
30. Bilal A, Alzahrani A, Almuhaimeed A, Khan AH, Ahmad Z, Long H. Advanced ckd detection through optimized metaheuristic modeling in healthcare informatics. Sci Rep. 2024;14(1):12601. https://doi.org/10.1038/s41598-024-63292-5.
31. Singh V, Asari VK, Rajasekaran R. A deep neural network for early detection and prediction of chronic kidney disease. Diagnostics. 2022;12(1):116. https://doi.org/10.3390/diagnostics12010116.
32. Saif D, Sarhan AM, Elshennawy NM. Early prediction of chronic kidney disease based on ensemble of deep learning models and optimizers. J Educ Chang Electr Syst Inf Technol. 2024;11(1). https://doi.org/10.1186/s43067-024-00142-4.
33. Chittora P, Chaurasia S, Chakrabarti P, Kumawat G, Chakrabarti T, Leonowicz Z, et al. Prediction of chronic kidney disease - a machine learning perspective. IEEE Access. 2021;9:17312–34. https://doi.org/10.1109/ACCESS.2021.3053763.
34. Schena FP, Anelli VW, Abbrescia DI, Noia TD. Prediction of chronic kidney disease and its progression by artificial intelligence algorithms. J Nephrol. 2022;35(8):1953–71. https://doi.org/10.1007/s40620-022-01302-3.
35. Rizwan M, Khan M, Gul M, Khan MA, Naseem R, Shaheen M. Comparative analysis of supervised machine learning algorithms for predicting chronic kidney diseases. In: Swaroop A, Virdee B, Correia SD, Valicek J, editors. Data processing and networking. Singapore: Springer; 2025. p. 451–63. https://doi.org/10.1007/978-981-96-3102-5_34.
36. Atalay M, Çelik E. The diagnosis and estimate of chronic kidney disease using the machine learning methods. ijisae. 2016;4(Special Issue-1):27–31. https://doi.org/10.18201/ijisae.265967.
37. Rubini. Chronic kidney disease. UCI Mach Learn Repository. 2015. https://doi.org/10.24432/C5G020.
38. Ekanayake IU, Herath D. Chronic kidney disease prediction using machine learning methods. 2020 Moratuwa Engineering Research Conference (MERCon). 2020, pp. 260–65). https://doi.org/10.1109/MERCon50084.2020.9185249.
39. Pasadana IA, Hartama D, Zarlis M, Sianipar AS, Munandar A, Baeha S, et al. Chronic kidney disease prediction by using different decision tree techniques. J Phys Conf Ser. 2019;1255(1):012024. https://doi.org/10.1088/1742-6596/1255/1/012024.
40. Devika R, Avilala SV, Subramaniyaswamy V. Comparative study of classifier for chronic kidney disease prediction using naive bayes, knn and random forest. 2019 3rd International Conference on Computing Methodologies and Communication (ICCMC). 2019, pp. 679–84). https://doi.org/10.1109/ICCMC.2019.8819654.
41. Khan B, Naseem R, Muhammad F, Abbas G, Kim S. An empirical evaluation of machine learning techniques for chronic kidney disease prophecy. IEEE Access. 2020;8:55012–22. https://doi.org/10.1109/ACCESS.2020.2981689.
42. Khan MA, Rizwan M, Bhatti SA, Khan M. Explainable ai-driven chronic kidney disease prediction system. 2025 IEEE International Conference on Emerging Trends in Engineering and Computing (ETECOM). 2025, pp. 1–6). https://doi.org/10.1109/ETECOM66111.2025.11319132.
43. Ghosh P, Shamrat FMJM, Shultana S, Afrin S, Anjum AA, Khan AA. Optimization of prediction method of chronic kidney disease using machine learning algorithm. 2020 15th International Joint Symposium on Artificial Intelligence and Natural Language Processing (iSAI-NLP). 2020, pp. 1–6). https://doi.org/10.1109/iSAI-NLP51646.2020.9376787.

44. Elhoseny M, Shankar K, Uthayakumar J. Intelligent diagnostic prediction and classification system for chronic kidney disease. Sci Rep. 2019;9(1):9583. https://doi.org/10.1038/s41598-019-46074-2.
45. Antony L, Azam S, Ignatious E, Quadir R, Beeravolu AR, Jonkman M, et al. F.D.: a comprehensive unsupervised framework for chronic kidney disease prediction. IEEE Access. 2021;9:126481–501. https://doi.org/10.1109/ACCESS.2021.3109168.
46. Krishnamurthy S, Ks K, Dovgan E, Lustrek M, Piletic BG, Srinivasan K, et al. Machine learning prediction models for chronic kidney disease using national health insurance claim data in Taiwan. Healthcare. 2021;9(5):546. https://doi.org/10.3390/healthcare9050546.
47. Lin L-Y, Warren-Gash C, Smeeth L, Chen P-C. Data resource profile: the national health insurance research database (NHIRD). Epidemiol Health. 2018;40:e2018062. https://doi.org/10.4178/epih.e2018062.
48. Ebiaredoh-Mienye SA, Swart TG, Esenogho E, Mienye ID. A machine learning method with filter-based feature selection for improved prediction of chronic kidney disease. Bioengineering. 2022;9(8):350. https://doi.org/10.3390/bioengineering9080350.
49. Moreno-Sanchez PA. Data-driven early diagnosis of chronic kidney disease: development and evaluation of an explainable ai model. IEEE Access. 2023;11:38359–69. https://doi.org/10.1109/ACCESS.2023.3264270.
50. Kumar A, Dhanka S, Sharma A, Sharma A, Maini S, Fahlevi M, et al. Comprehensive framework for thyroid disorder diagnosis: integrating advanced feature selection, genetic algorithms, and machine learning for enhanced accuracy and other performance matrices. PLoS One. 2025;20(6):1–25. https://doi.org/10.1371/journal.pone.0325900.
51. Singh J, Pandey B, Karna S, Arora AS, Kumar A. Enhancing the thermographic diagnosis of maxillary sinusitis using deep learning approach. Quant Infrared Thermogr J. 2025;22(3):195–209. https://doi.org/10.1080/17686733.2024.2349976.
52. Arif MS, Rehman AU, Asif D. Explainable machine learning model for chronic kidney disease prediction. Algorithms. 2024;17(10):443. https://doi.org/10.3390/a17100443.
53. Rizwan M, Naseem R. A benchmark biomarker dataset for classification of chronic kidney disease patients in Pakistan. Figshare. 2025. https://doi.org/10.6084/m9.figshare.29603663.
54. Rahman M, Al-Amin M, Hossain J. Machine learning models for chronic kidney disease diagnosis and prediction. Biomed Signal Process Control. 2024;87:105368. https://doi.org/10.1016/j.bspc.2023.105368.
55. Dharmarathne G, Bogahawaththa M, McAfee M, Rathnayake U, Meddage DPP. On the diagnosis of chronic kidney disease using a machine learning-based interface with explainable artificial intelligence. Intell Syst Appl. 2024;22:200397. https://doi.org/10.1016/j.iswa.2024.200397.
56. Ghosh SK, Khandoker AH. Investigation on explainable machine learning models to predict chronic kidney diseases. Sci Rep. 2024;14(1):3687. https://doi.org/10.1038/s41598-024-54375-4.

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